q_{en}/ϵ_0

$\vec{B} = \left(\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j}\right)$ $\left[\omega\left(t - \frac{x}{C}\right)\right]$

$\vec{E} = \left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right)$ $\left(\omega t - \frac{\omega x}{C}\right)$

$\vec{E} = \left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right)$ $\left(\omega t - \frac{\omega x}{C}\right)$

$\vec{E} = \left(\frac{1}{2}\hat{i} + \frac{\sqrt{3}}{2}\hat{j}\right)$ $\left(\omega t - \frac{\omega x}{C}\right)$

$\vec{E} = \left(\frac{1}{2}\hat{i} - \frac{\sqrt{3}}{2}\hat{j}\right)$ $\left(\omega t - \frac{\omega x}{C}\right)$

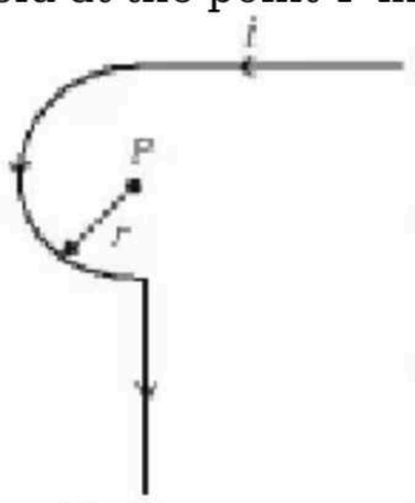
$$\vec{E}_1 = E_0\hat{j}\cos(\omega t - kx) \qquad \vec{E}_2 = E_0\hat{k}\cos(\omega t - ky)$$

$$\vec{v} = 0.8c\hat{j}$$

(3) 1 : 3

(4) $1 : \sqrt{3}$

13) Find the magnetic field at the point P in figure. The curved portion is a semicircle connected to



two long straight wires.

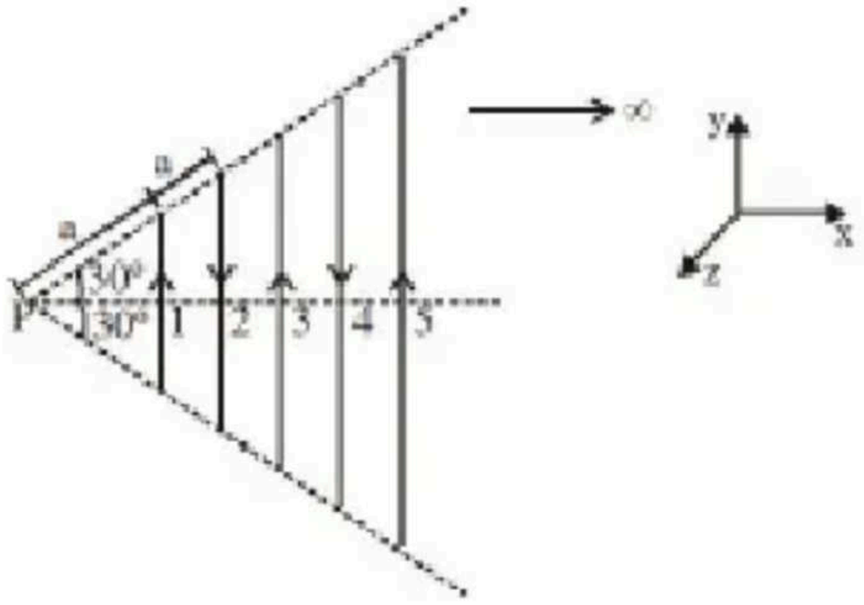
(1) $\frac{\mu_0 i}{2r}\left(1+\frac{2}{\pi}\right)$

(2) $\frac{\mu_0 i}{2r}\left(1+\frac{1}{\pi}\right)$

(3) $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{2\pi}\right)$

(4) $\frac{\mu_0 i}{2r}\left(\frac{1}{2}+\frac{1}{\pi}\right)$

14) Infinite number of straight wires each carrying current I are equally placed as shown in the figure. Adjacent wires have current in opposite direction. Net magnetic field at point P is :-



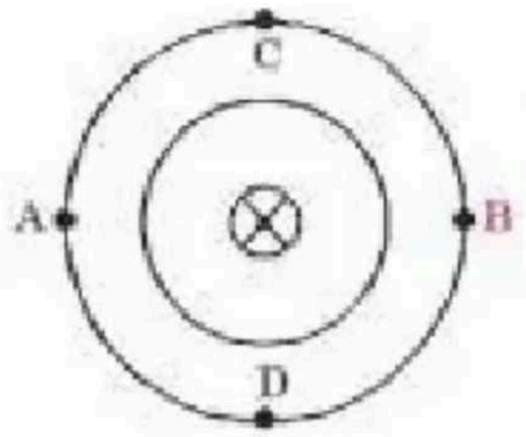
(1) $\frac{\mu_0 I \ell n 2}{4\pi\ \sqrt{3}a}\hat{k}$

(2) $\frac{\mu_0 I \ell n 4}{4\pi\ (\sqrt{3})a}\hat{k}$

(3) $\frac{\mu_0 I \ell n 4}{4\pi\ \sqrt{3}a}(-\hat{k})$

(4) Zero

15) Fig. shows two magnetic field lines of a long conductor, located perpendicular to the plane of the paper carrying inward current. In which one of the points A, B, C, D direction of the magnetic field



(B) is towards right

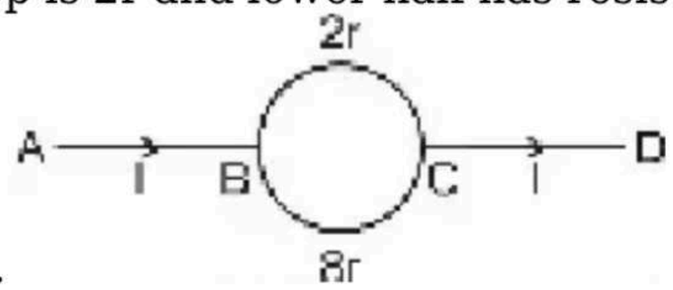
(1) A

(2) B

(3) C

(4) D

16) If current I enters the loop through wire AB and leaves the loop through point C. if the resistance of upper half of loop is 2r and lower half has resistance 8r. Find out value of magnetic induction if loop has radius R.



(1) $\frac{3\mu_0 I}{20R}$ inwards

(2) $\frac{3\mu_0 I}{20R}$ outwards

(3) $\frac{7\mu_0 I}{20R}$ inwards

(4) $\frac{7\mu_0 I}{20R}$ outwards

17) The magnetic field at the centre of a circular coil of radius r, due to current I flowing through it, is B. The magnetic field at a point along the axis at a distance $\frac{r}{2}$ from the centre is :

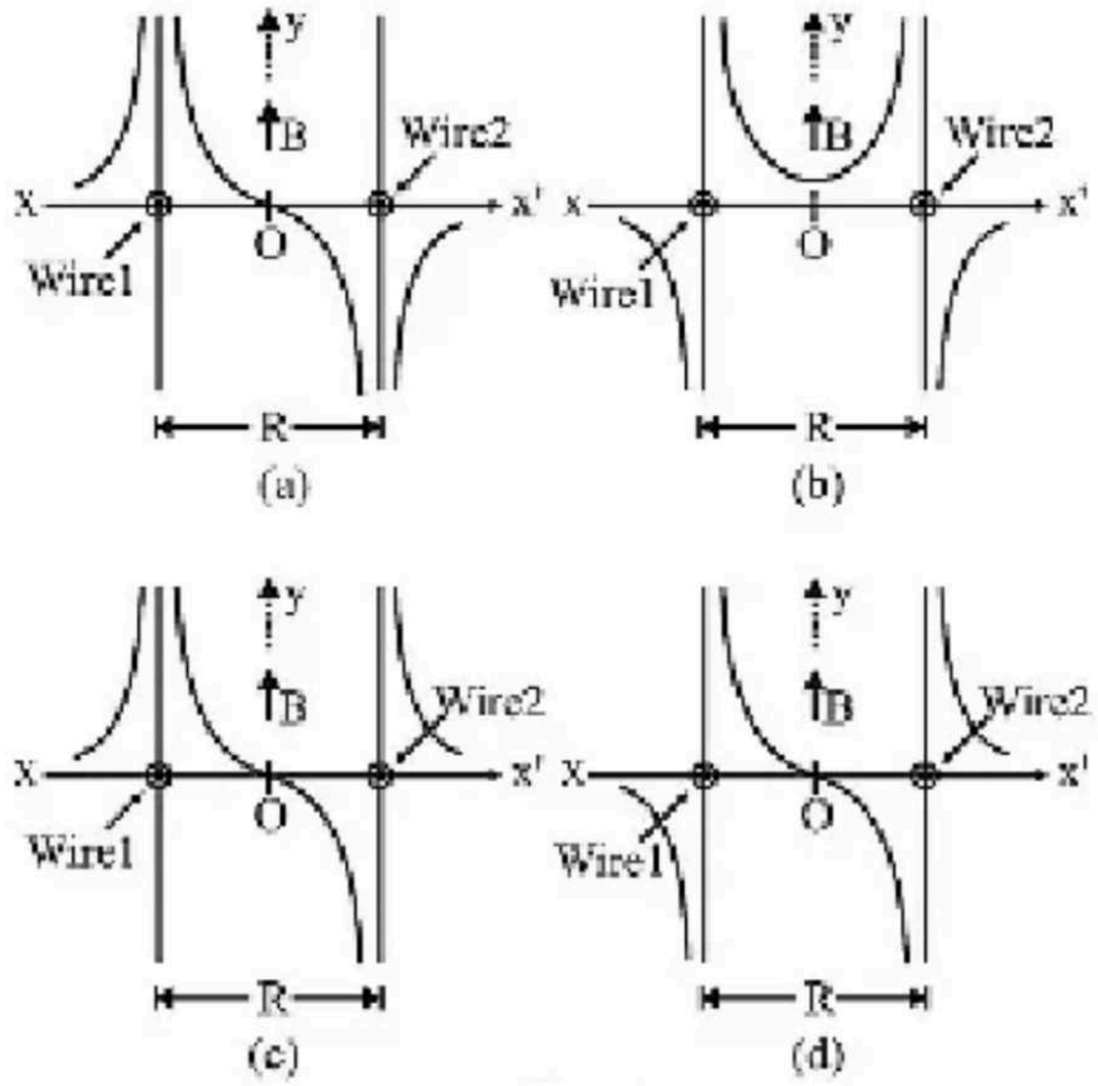
(1) B/2

(2) 2B

(3) $\left(\frac{2}{\sqrt{5}}\right)^3 B$

(4) $\left(\frac{2}{\sqrt{3}}\right)^3 B$

18) Two long parallel wires are at a distance R apart. They carry steady equal current flowing out of the plane of the paper, as shown in the following figures. The variation of the magnetic field B along



the line xx' is given by :-

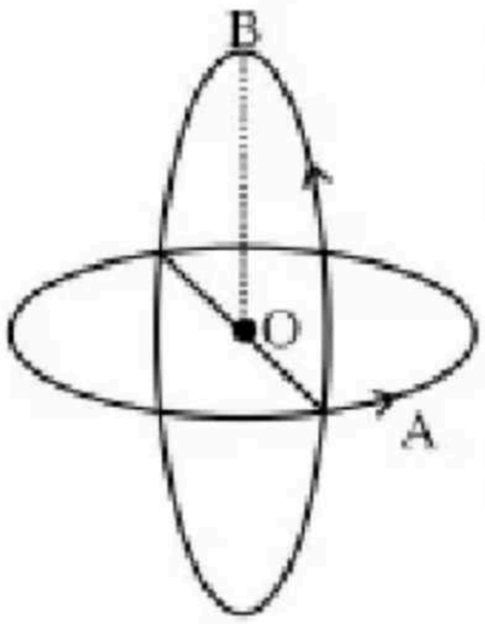
(1) a

(2) b

(3) c

(4) d

19) Two insulated circular loop A and B radius ‘a’ carrying a current of ‘I’ in the anti clockwise direction as shown in figure. The magnitude of the magnetic induction at the centre will be :



(1) $\frac{\sqrt{2}\mu_0 I}{a}$

(2) $\frac{\mu_0 I}{2a}$

(3) $\frac{\mu_0 I}{\sqrt{2}a}$

(4) $\frac{2\mu_0 I}{a}$

20) Due to the flow of current in a circular loop of radius R, the magnetic induction produced at the centre of the loop is B. The manetic moment of the loop is [μ_0 = Permeability constant] :-

(1) $\frac{BR^3}{2\pi\mu_0}$

(2) $\frac{2\pi BR^3}{\mu_0}$

(3) $\frac{BR^2}{2\pi\mu_0}$

(4) $\frac{2\pi BR^2}{\mu_0}$

21) To find the resistance of galvanometer by half deflection method, experimental data obtained is given in table below :

S. No.	Resistance R (Ω)	Deflection in galvanometer (θ)	Shunt resistance S (Ω)	Half deflection (θ/2)	Galvanometer resistance (G)
1	3300	30	80	15	G_1
2	5000	20	80	10	G_2

From the above data, the galvanometer resistance R will be near to :

(1) 20Ω

(2) 40Ω

(3) 50Ω

(4) 80Ω

22) In the half-deflection method, the galvanometer is connected in series with :

(1) High resistance

(2) Capacitor

(3) Voltmeter

(4) Ammeter

23) A long straight wire in the horizontal plane carries a current of 50 A in north to south direction. Give the magnitude and direction of B at a point 2.5m east of the wire :-

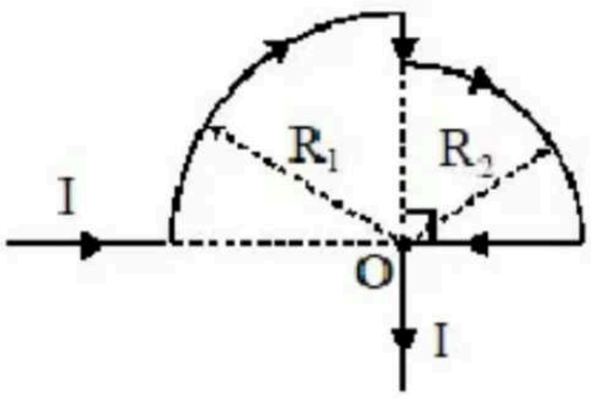
(1) 4×10^{-6} T upwards

(2) 6×10^{-6} T down

(3) 4×10^{-6} T east

(4) Both (1) and (3)

24) In the loop shown, the magnetic induction at the point 'O' is



(1) $\frac{\mu_0 I}{8}\left(\frac{R_1 - R_2}{R_1 R_2}\right)$

(2) $\frac{\mu_0 I}{8}\left(\frac{R_1 + R_2}{R_1 R_2}\right)$

(3) $\frac{\mu_0 I}{8}\left(\frac{R_1 R_2}{R_1 + R_2}\right)$

(4) Zero

25) A hollow cylindrical wire carries current I, having inner & outer radius R & 2R respectively magnetic field at a point which is $\frac{5R}{4}$ distance away from the axis of wire :-

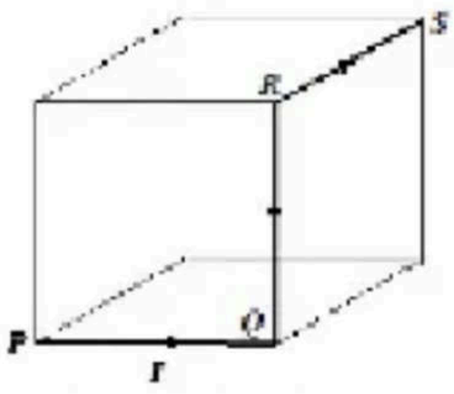
(1) $\frac{5\mu_0 I}{18\pi R}$

(2) $\frac{\mu_0 I}{36\pi R}$

(3) $\frac{5\mu_0 I}{36\pi R}$

(4) $\frac{3}{40}\frac{\mu_0 I}{\pi R}$

26) A wire PQRS runs along three edges of a cube of side □ as shown carrying a current I. There exists a uniform magnetic field of magnitude B along one of the sides of the cube. The magnitude of the force acting on the wire is



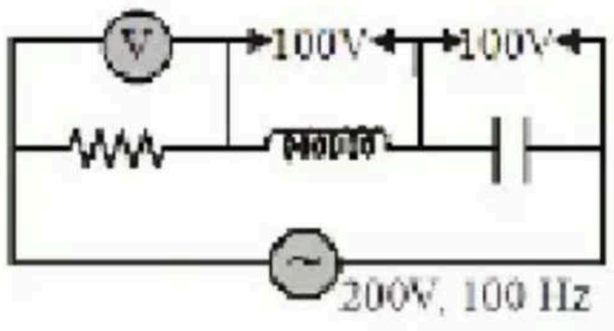
(1) 0

(2) I□B

(3) $\sqrt{2}I\ell B$

(4) $2I\ell B$

27) In the circuit given below, what will be the reading of the voltmeter :-

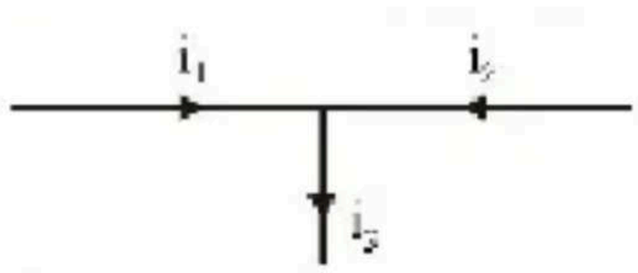


(1) 300 V

(2) 900 V

(3) 200 V

(4) 400 V



28) In the given figure if $i_1 = 3 \sin \omega t$ and $i_2 = 4 \cos \omega t$, then i_3 is

(1) $5 \sin (\omega t + 53°)$

(2) $5 \sin (\omega t + 37°)$

(3) $5 \sin (\omega t + 45°)$

(4) $5 \cos (\omega t + 53°)$

29) The frequencies at which the current amplitude in an LCR series circuit becomes $\frac{1}{\sqrt{2}}$ times its maximum value, are 212 rad s^{-1} and 232 rad s^{-1}. The value of resistance in the circuit is $R = 5\Omega$. The self inductance in the circuit is _____ mH.

(1) 200 mH

(2) 250 mH

(3) 500 mH

(4) 100 mH

30) The rms value of conduction current in a parallel plate capacitor is 6.9 μA. The capacity of this capacitor, if it is connected to 230 V ac supply with an angular frequency of 600 rad/s, will be :

(1) 5 pF

(2) 50 pF

(3) 100 pF

(4) 200 pF

31) Given below are two statements :
Statement-I **:** The reactance of an ac circuit is zero. It is possible that the circuit contains a capacitor and an inductor.
Statement-II : In ac circuit, the average power delivered by the source never becomes zero.

(1) Both Statement I and Statement II are true.

(2) Both Statement I and Statement II are false.

(3) Statement I is true but Statement II is false.

(4) Statement I is false but Statement II is true.

32) The capacitance of a telephone wire of length 300 km is 0.01 μF per km. If wire caries an ac of 5 kHz, what should be the value of an inductance required to be connected in series so that impedance is minimum-

(1) 0.34 mH

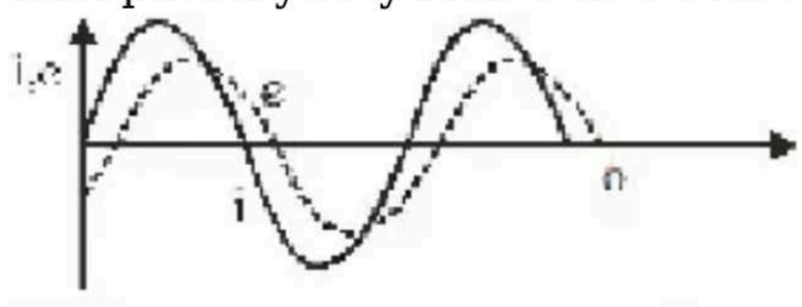
(2) 3.6 mH

(3) 0.6 H

(4) 3.6 H

33) When an AC source of e.m.f. $e = E_0 \sin(100t - \pi/4)$ is connected across a circuit, the phase difference between the e.m.f. e and the current i in the circuit is observed to be $\frac{\pi}{4}$, as shown in the diagram.

If the circuit consists possibly only of R–C or R–L or L–C in series, find the relationship between the two elements :–



(1) R = 1 kΩ, C = 10 μF

(2) R = 1 kΩ, C = 1 μF

(3) R = 1 kΩ, L = 10 H

(4) R = 1 kΩ, L = 1 H

34) If a choke coil of negligble resistance works on 220 volt source and 5 mA current is flowing through it, then the loss of power in choke coil is:

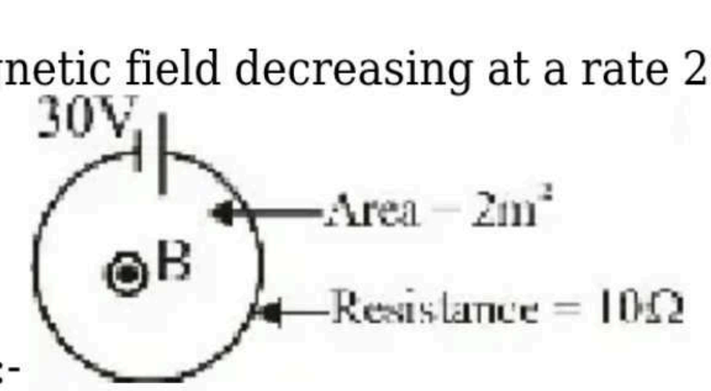
(1) zero

(2) 11 watt

(3) 44×10^3 watt

(4) 1.1 watt

35) Figure shows a coil placed in a magnetic field decreasing at a rate 20 T/Sec. A battery of 30 V is also connected then current in the coil :-



(1) 7A, Clockwise

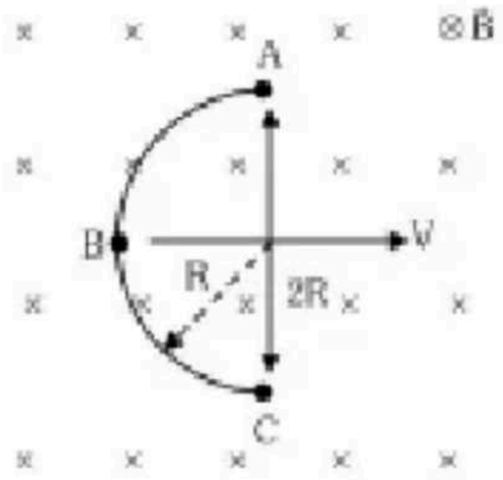
(2) 7 A, Anticlockwise

(3) 1 A, Clockwise

(4) 1 A, Anticlockwise

36)

A circular arc is moving in a uniform magnetic field as shown, Emf induced across AB is -



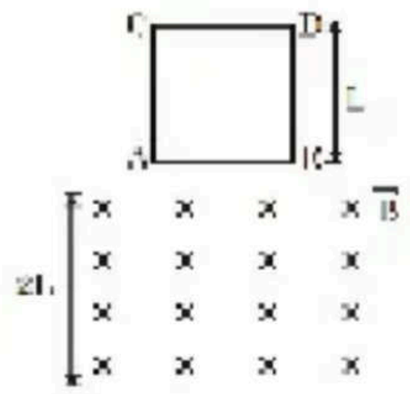
(1) BvR

(2) 2BvR

(3) $\sqrt{2}$BvR

(4) None of these

37) A square coil ACDE with its plane vertical is released from rest in a horizontal uniform magnetic



$\vec{B}$ of length 2L. The acceleration of the coil is

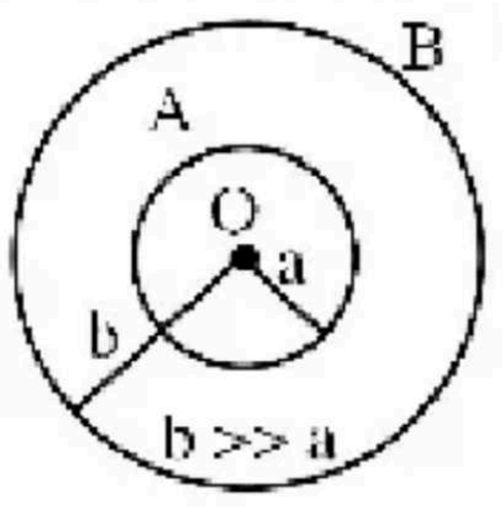
(1) Less than g for all the time till the loop crosses the magnetic field completely

(2) Less than g when it enters the field and greater than g when it comes out of the field

(3) Equal to g when it is completely within the field

(4) g all the time

38) Two conducting circular loops A and B are placed in the same plane with their centres



coinciding as shown in figure. The mutual inductance between them is:

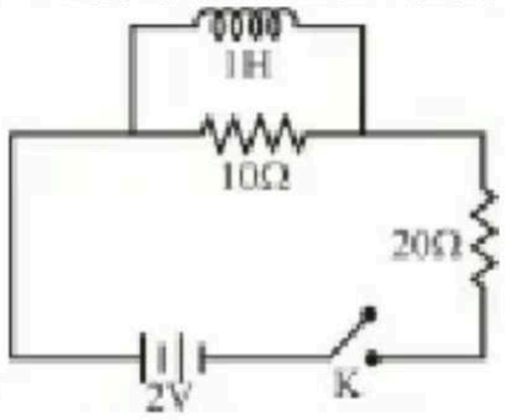
(1) $\frac{\mu_0 \pi a^2}{2b}$

(2) $\frac{\mu_0}{2\pi} \cdot \frac{b^2}{a}$

(3) $\frac{\mu_0 \pi b^2}{2a}$

(4) $\frac{\mu_0}{2\pi} \cdot \frac{a^2}{b}$

39) In the adjoining figure what is the final value of current in the 10 ohm resistor long after the plug of key K is inserted ?



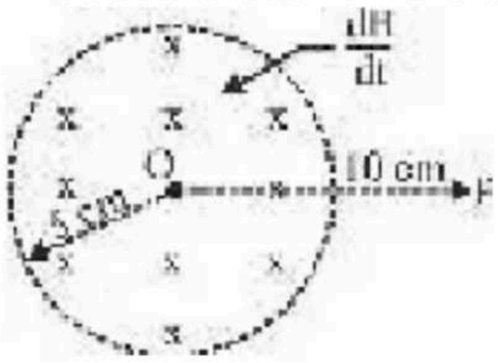
(1) (3/10) A

(2) (3/20) A

(3) (3/11) A

(4) Zero

40) Magnetic fleld changing at the rate of 4T/sec in a circular region of 5 cm radius. The value of induced electric field at a point 'P', which is 10 cm distance away from the centre 'O' of region is :-



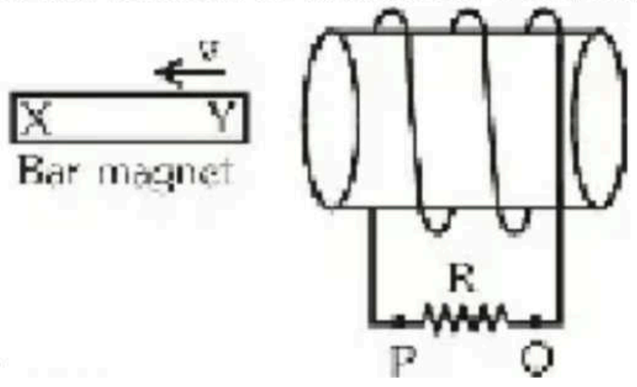
(1) 0.05 V/m

(2) 0.2 V/m

(3) 0.5 V/m

(4) 2V/m

41) A bar magnet moves away from the solenoid along its axis, then direction of induced current in resistance 'R' is from P to Q then find out polarity of bar magnet :-



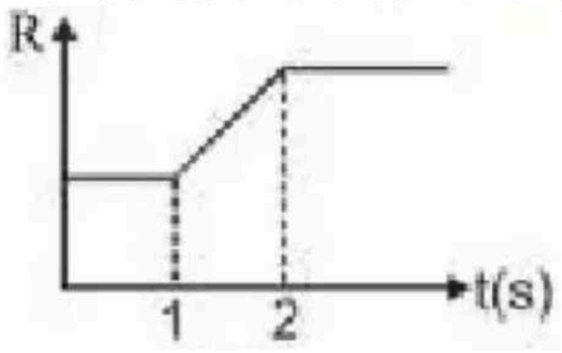
(1) X → North pole, Y → South pole

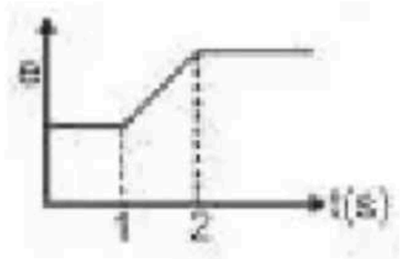
(2) X → South pole, Y → North pole

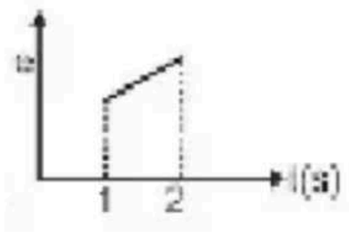
(3) Impossible

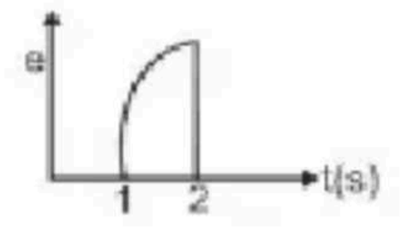
(4) None of these

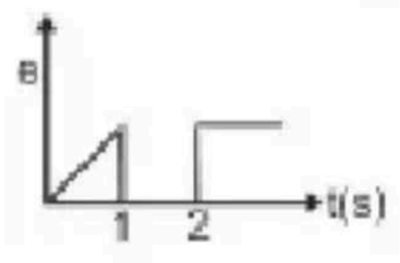
42) A flexible wire bent in the form of a circular coil is placed in a uniform magnetic field perpendicular to the plane of the coil. The radius of the coil changes with time as shown in the figure. The induced emf in the coil is :–



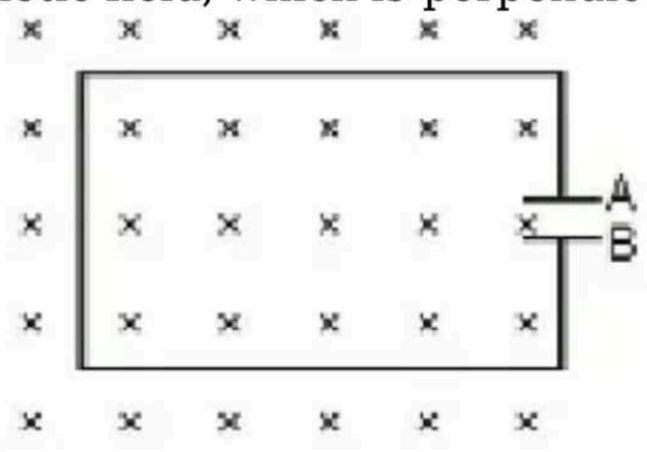
(1) 

(2) 

(3) 

(4) 

43) In the given figure if the magnetic field, which is perpendicular to the plane of the paper in the inward direction increases, then-

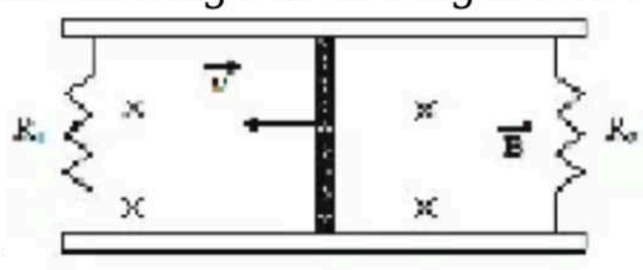


(1) Plate B of the capacitor will become positively charged.

(2) Plate A of the capacitor will become positively charged.

(3) The capacitor will not be charged.

(4) Both plates will be charged alternately.

44) A conducting bar of length L is free to slide on two parallel conducting rails as shown in the figure.  Two resistors R_1 and R_2 are connected across the ends of the rails. There is a uniform magnetic field $\vec{B}$ Perpendicular inward to the plane of paper. An external agent pulls this rod with constant speed v towards left. Then which option is correct regarding I_1 and I_2 currents flowing through R_1 and R_2 respectively is :

(1) Both I_1 and I_2 are in anticlockwise direction

(2) Both I_1 and I_2 are in clockwise direction

(3) I_1 is in clockwise direction and I_2 is in anticlockwise direction

(4) I_1 is in anticlockwise direction and I_2 is in clockwise direction

45) The source of time varying magnetic field may be
(A) a permanent magnet at rest

(B) an electric field changing linearly with time
(C) direct current
(D) a decelerating charge particle
Choose the correct answer from the options given below:

(1) (D) only

(2) (B) and (C) only

(3) (A) only

(4) (B) and (D) only

CHEMISTRY

1) Evaluate the following statements related to group 14 elements for their correctness.
(A) Covalent radius decreases down the group from C to Pb in a regular manner.
(B) Electronegativity decreases from C to Pb down the group gradually.
(C) Maximum covalence of C is 4 whereas other elements can expand their covalence due to presence of d orbitals.
(D) Heavier elements do not form pπ-pπ bonds.
(E) Carbon can exhibit negative oxidation states.
Choose the **correct** answer from the options given below :

(1) (C), (D) and (E) Only

(2) (A) and (B) Only

(3) (A), (B) and (C) Only

(4) (C) and (D) Only

2) Which of the following statement is **INCORRECT**?

(1) The melting point of hydrides of Group 15 : $NH_3 > SbH_3 > AsH_3 > PH_3$

(2) ΔH_{eg} kJ mol^{-1} : S > Se > O > Te > Po

(3) Melting point : B > A□ > T□ > In > Ga

(4) Bond enthalpy : C – C > Si – Si > Ge – Ge > Sn – Sn

3) Some compounds are given below :-
MnF_7, $PbBr_4$, PbI_4, BiF_5, $BiCl_5$, T□(I)I_3, PbO_2
How many compounds do not exist.

(1) Three

(2) Two

(3) Five

(4) Four

4)

	Column-I		**Column-II**
(A)	B_2H_6	(P)	Non-planar

(B)	Al_2Cl_6	(Q)	sp^3 hybridised (anyone atom)
(C)	I_2Cl_6	(R)	$3c–2e^-$ or $3c–4e^-$ bond is present
(D)	Si_2H_6	(S)	Vacant orbital participate in hybridisation
		(T)	X–X bond is present (X = central atom)

Choose the correct matching in above given column.

(1) A→P,Q,R,S; B→P,Q,R,S; C→Q,R,S; D→P,Q,T

(2) A→P,Q,R,T; B→Q,R,S,T; C→Q,T; D→P,Q,T

(3) A→P,Q,R,S; B→P,Q,R; C→Q,R,S; D→P,Q,T

(4) A→P,Q,R; B→P,Q,R; C→Q,R,S; D→P,Q,T

5) If x = Number of P – OH bond
y = Number of P – H bond
z = Number of P = O bond
Then the value of 2x + y – z in phosphonic acid ?

(1) 6

(2) 5

(3) 4

(4) 3

6) Find the number of molecules, when they undergo hydrolysis at room temperature the produced acid from central atom has the basicity of '2'.
(PCl_5, SF_6, SF_4, P_4O_6, PCl_3)

(1) 4

(2) 3

(3) 2

(4) 5

7) Which statement about white phosphorous is correct ?

(1) White phosphorous contain 60° bond angle

(2) It dissolve in boiling NaOH solution in an inert atmosphere giving PH_3

(3) It readily catches fire to give dense white fumes at P_4O_{10}

(4) All of these

8) **Assertion :** Compounds of boron are covalent in nature.
Reason : Due to small size of boron, the sum of its first three ionisation enthalpies is very high to prevent it to form +3 ions.

(1) Assertion and Reason both are True and Reason is correct explanation of Assertion.

(2) Both Assertion and Reason are True and Reason is not correct explanation of Assertion.

(3) If Assertion is True and Reason is False.

(4) If Assertion is False and Reason is True.

9) Choose correct statement.
(A) Polonium occurs in nature as a decay product of thorium and uranium minerals.
(B) Oxygen is the most abundant among all the elements on earth.
(C) Polonium does not show '–2' oxidation state.

(1) Only A

(2) Only B

(3) Only C

(4) A, B and C

10) Identify the incorrect option from the following :

(1) Fluoroapatite –$3Ca_3(PO_4)_2.CaF_2$

(2) Carnallite – KCl. $MgCl_2.6H_2O$

(3) Cryolite – Na_3AlF_6

(4) Fluorspar – BF_3

11) Incorrect order of properties of noble gas.

(1) Boiling point → He < Ne < Ar < Kr < Xe

(2) Ionisation energy → He > Ne > Ar > Kr > Xe

(3) Density → He < Ne < Ar < Kr < Xe

(4) ΔH_{eg} (in +ve) → Ne > He > Ar > Kr > Xe

12) Compound of radon have not been isolated but only identified by radiotracer technique, the compound is -

(1) RnF_6

(2) RnF_4

(3) RnF_2

(4) RnF

13) Electronic Configuration of Th^{+3} is

(1) $[Rn]6d^1$

(2) $[Rn]5f^1$

(3) $[Rn]7s^1$

(4) None of these

14) The spin only magnetic moment value of the ion among Ti^{2+}, V^{2+}, Co^{3+} and Mn^{+3}, that acts as strongest oxidising agent in aqueous solution is ..... BM.

(Given atomic numbers : Ti : 22, V : 23, Cr : 24, Co : 27)

(1) $\sqrt{8}$

(2) $\sqrt{15}$

(3) $\sqrt{24}$

(4) $\sqrt{35}$

15) Which of the following statements is/are correct ?

(i) Orange solution of sodium dichromate, can be crystallised as $Na_2Cr_2O_7.2H_2O$.

(ii) Cu^{+} ion disproportionate in aqueous solution into Cu and Cu^{3+}

(iii) Cr^{+2} can liberate hydrogen gas from a dilute acid

(iv) Cr_2O_3 and MnO_2 are amphoteric oxides

(v) Sc and Zn both are transition metals

(1) i, ii, iv, v

(2) i, iii, v

(3) i, iii, iv

(4) ii, iii, iv

16) Match the following group of metals with their characteristic in d-block

Column-I		**Column-II**	
(A)	Cr, Mo,W	(i)	Maximum oxidation state
(B)	Ru, Os	(ii)	Strong metallic bond
(C)	Cu,Ag, Au	(iii)	Volatile metals
(D)	Zn, Cd, Hg	(iv)	Coinage metals

(1) A-(ii), B-(i), C-(iii), D-(iv)

(2) A-(ii), B-(i), C-(iv), D-(iii)

(3) A-(i), B-(iv), C-(ii), D-(iii)

(4) A-(iii), B-(ii), C-(i), D-(iv)

17) Choose the correct option for the following statements -

Statement-I : In general the electronic configuration of outer orbitals of d-block elements is $(n–1)d^{1–10}\ ns^{1–2}$ except Pd.

Statement-II : The (n–1) stands for the inner d-orbital which may have one to ten electrons & the outermost ns orbital may have one or two electrons.

(1) Both Statement-I and Statement-II are correct.

(2) Statement-I is correct but Statement-II is incorrect.

(3) Statement-I is incorrect but Statement-II is correct.

(4) Both Statement-I and Statement-II are incorrect.

18)

Which of the following statement is not correct?

(1) $La(OH)_3$ is less basic than $Lu(OH)_3$

(2) In lanthanide series, ionic radius of Ln^{3+} ions decreases

(3) La is actually an element of transition series rather than lanthanide series

(4) Atomic radii of Zr and Hf are same because of lanthanide contraction.

19) Match the following compounds with their uses :

(A)	MnO_2	(i)	Manufacture of polythene
(B)	V_2O_5	(ii)	Dry battery cells
(C)	$TiC\square_4$ + $A\square(CH_3)_3$	(iii)	Used in formation of H_2SO_4
(D)	AgBr	(iv)	Photographic material

(1) A(ii), B(i), C(iii), D(iv)

(2) A(ii), B(iii), C(iv), D(i)

(3) A(ii), B(iii), C(i), D(iv)

(4) A(iii), B(ii), C(i), D(iv)

20) Which one of the following statements is **INCORRECT**?

(1) The actinoids show a greater range of oxidation states than the lanthanoids because the 5f, 6d and 7s levels are of comparable energies.

(2) 5f electrons can participate in bonding to a far greater extent than that 4f orbitals because unlike 4f orbitals, 5f orbitals are not as buried as 4f orbitals.

(3) Lanthanoid contraction is more important than the actinoid contraction because the chemistry of elements succeeding the actinoids are much less known at the present time.

(4) The magnetic properties of the actinoids are less complex than the lanthanoids.

21) Number of cis type of isomers for a square planar complex of type Mabcd.

(1) 0

(2) 1

(3) 3

(4) 2

22) Which of the following statement(s) is/are correct regarding complex ion $[Co(en)_2(NH_3)_2]^{3+}$

(I) Central metal is sp^3d^2 hybridised

(II) Complex ion shows geometrical as well as optical isomerism.

(III) It has two N–Co–N bond angles.

(1) I & II

(2) II & III

(3) Only II

(4) Only III

23) Increasing order of electrical conductivity of following complex compound is -

I. $[Pt(NH_3)_6]C\square_4$
II. $[Cr(NH_3)_6]C\square_3$
III. $[Co(NH_3)_4C\square_2]C\square$
IV. $K_2[PtC\square_6]$

(1) III < IV < II < I

(2) IV < II < III < I

(3) II < I < IV < III

(4) I < II < IV < III

24) Match the following complex with their magnetic moment

Column-I		**Column-II**	
(A)	$[Fe(CN)_6]^{-4}$	(i)	1.73
(B)	$[Fe(CN)_6]^{+3}$	(ii)	0.00
(C)	$[Cr(H_2O)_6]^{+3}$	(iii)	2.83
(D)	$[Ni(H_2O)_6]^{+2}$	(iv)	3.88

(1) A-(ii), B-(iv), C-(i), D-(iii)

(2) A-(ii), B-(i), C-(iv), D-(iii)

(3) A-(i), B-(iii), C-(ii), D-(iv)

(4) A-(ii), B-(iii), C-(i), D-(iv)

25) No. of metal ions among the following which has no effect of strong field or weak field ligand on their magnetic character in octahedral complex.
Ni^{+2}, Pt^{+4}, Co^{+3}, Cr^{+3}, Fe^{+3}

(1) 3

(2) 4

(3) 5

(4) 2

26) Given below are two statements:
Statement (I): A solution of $[Fe\ (H_2O)_6]^{2+}$ is green in colour.
Statement (II): A solution of $[Ni(CN)_4]^{2-}$ is colourless.
choose the most appropriate answer from the options given below:

(1) Both Statement I and Statement II are incorrect

(2) Both Statement I and Statement II are correct

(3) Statement I is incorrect but Statement II is correct

(4) Statement I is correct but Statement II is incorrect

27) For a d^4 metal ion in an octahedral field, the correct electronic configuration is :

(1) $t_{2g}^4 e_g^0$ when $\Delta_O < P$

(2) $e_g^2 t_{2g}^2$ when $\Delta_O < P$

(3) $t_{2g}^3 e_g^1$ when $\Delta_O < P$

(4) $t_{2g}^3 e_g^1$ when $\Delta_O > P$

28) Match List I with List II

LIST-I (Substances)		**LIST-II (Element Present)**	
(A)	Ziegler catalyst	(I)	Rhodium
(B)	Blood Pigment	(II)	Cobalt
(C)	Wilkinson catalyst	(III)	Iron
(D)	Vitamin B_{12}	(IV)	Titanium

Choose the correct answer from the options given below:

(1) A-II, B-IV, C-I, D-III

(2) A-II, B-III, C-IV, D-I

(3) A-III, B-II, C-IV, D-I

(4) A-IV, B-III, C-I, D-II

29) The correct IUPAC name of K_2MnO_4 is

(1) Potassium tetraperoxopermanganate (VI)

(2) Potassium tetraoxidomanganate (VI)

(3) Dipotassium tetraoxidomanganate (VII)

(4) Potassium tetraoxidomanganese (VI)

30) Match List-I with List-II.

LIST – I (Complex ion)		**LIST – II (Electronic Configuration)**	
A.	$[Cr(H_2O)_6]^{3+}$	I.	$t_{2g}^2 e_g^0$
B.	$[Fe(H_2O)_6]^{3+}$	II.	$t_{2g}^3 e_g^0$
C.	$[Ni(H_2O)_6]^{2+}$	III.	$t_{2g}^3 e_g^2$
D.	$[V(H_2O)_6]^{3+}$	IV.	$t_{2g}^6 e_g^2$

Choose the correct answer from the options given below :

(1) A-III, B-II, C-IV, D-I

(2) A-IV, B-I, C-II, D-III

(3) A-IV, B-III, C-I, D-II

(4) A-II, B-III, C-IV, D-I

31) Given below are two statements :

Statement (I) : Dimethyl glyoxime forms one six-membered covalent chelate when treated with $NiCl_2$ solution in presence of NH_4OH.
Statement (II) : Prussian blue precipitate contains iron both in (+2) and (+3) oxidation states.
Choose the most appropriate answer from the options given below :

(1) Statement I is false but Statement II is true

(2) Both Statement I and Statement II are true

(3) Both Statement I and Statement II are false

(4) Statement I is true but Statement II is false

32) Given below are two statements:
Statement-I: The gas liberated on warming a salt with dil. H_2SO_4, turns a piece of paper dipped in lead acetate into black, it is a confirmatory test for sulphide ion.
Statement-II: In statement-I the colour of paper turns black because of formation of lead sulphite.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both Statement-I and Statement-II are false

(2) Statement-I is false but Statement-II is true

(3) Statement-I is true but Statement-II is false

(4) Both Statement-I and Statement-II are true

33) Number of metal ions characterized by flame test among the following is _____.
$Sr^{2+}, Ba^{2+}, Ca^{2+}, Cu^{2+}, Zn^{2+}, A\ell^{+3}, Sn^{+2}$

(1) 4

(2) 5

(3) 6

(4) 7

34) When ethane – 1, 2- diamine is added progressively to an aqueous solution of Nickel (II) chloride the sequence of color change observed will be : -

(1) Pale Blue → Blue → Green → Violet

(2) Pale Blue → Blue → Violet → Green

(3) Green → Pale Blue → Blue → Violet

(4) Violet → Blue → Pale Blue → Green

35) On passing a gas, 'X', through Nessler's reagent, a brown precipitate is obtained. The gas 'X' is

(1) H_2S

(2) CO_2

(3) NH_3

(4) Cl_2

36)

Identify the correct order of solubility of Na_2S, CuS and ZnS in aqueous medium :-

(1) CuS > ZnS > Na_2S

(2) ZnS > Na_2S < CuS

(3) Na_2S > CuS > ZnS

(4) Na_2S > ZnS > CuS

37) Sometimes yellow turbidity appears while passing H_2S gas even in the absence of second group radicals. This is because :-

(1) Sulphur is present in the mixture as an impurity

(2) The fourth group radicals get precipitated as sulphides

(3) Of the oxidation of H_2S gas by some radicals acting as oxidant

(4) The third group radicals get precipitated as hydroxides

38) Al^{3+}, Fe^{3+} and Cr^{3+} are grouped together for qualitative analysis because their :-

(1) Carbonates are insoluble in NH_3

(2) Hydroxides are insoluble in NH_3

(3) Sulphides are soluble in acid

(4) None of these

39) A brick red colour is imparted to Bunsen flame by a :

(1) Ca salt

(2) Sr salt

(3) Na salt

(4) Co salt

40) Which one of the following ions does not give borax bead test ?

(1) Cr^{3+}

(2) Cu^{2+}

(3) Mn^{2+}

(4) Zn^{2+}

41) The stability constants of the complexes formed by a metal ion (M^{2+}) with NH_3, CN^-, H_2O and en are of the order 10^{11}, 10^{27}, 10^{15} and 10^8 respectively. Then :-

(1) en is the strongest ligand

(2) CN^- is the strongest ligand

(3) The strength of the ligands has no relationship with given values

(4) All ligands are equally strong

42) Which one of the following pairs of isomers and types of isomerism are correctly matched :-
(a) $[Co(NH_3)_5(NO_2)]Cl_2$ and

$[Co(NH_3)_5(ONO)]Cl_2$.....Linkage

(b) $[Cu(NH_3)_4]$ $[PtCl_4]$ and

$[Pt(NH_3)_4][CuCl_4]$.....co-ordination

(c) $[Pt(NH_3)_4Cl_2]Br_2$ and

$[Pt(NH_3)_4Br_2)]Cl_2$.....Ionization

Select the correct answer using the codes given below :-

(1) (b) and (c)

(2) (a), (b) and (c)

(3) (a) and (c)

(4) (a) and (b)

43) The geometry of some complex ions are given against them :-

(A) $[Ag(NH_3)_2]^+$ – Linear

(B) $[MnCl_4]^{2-}$ – Tetrahedral

(C) $[Cu(NH_3)_4]^{2+}$ – Square planar

(D) $[Ni(CN)_4]^{2-}$ – Square planar

The correct match is :-

(1) A and D

(2) A, B and C

(3) A, C and D

(4) A, B, C and D

44) The secondary valency and the number of hydrogen bonded water molecule(s) in $CuSO_4.5H_2O$, respectively, are

(1) 6, 4

(2) 4, 1

(3) 6, 5

(4) 5, 1

45) For the complex ion, dichloridobis-(ethylenediamine) cobalt(III), select the correct statement.

(1) It has three isomers, two of them are optically active and one is optically inactive

(2) It has three isomers, all of them are optically active

(3) It has three isomers, all of them are optically inactive

(4) It has only one optically active isomer and three geometrical isomers

BIOLOGY

1) Given belows are two statements :

Statement-I :- In AB blood group individual I^A and I^B are present together and both express their own types of sugars this is because of co-dominance hence Red blood cells have both A and B types of sugar.

Statement-II :- In ABO blood group 4 different genotype and 6 different phenotypes are possible.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I & statement II are incorrect

(4) Both statement I & statement II are correct

2) **Assertion (A) :-** If a gene controls more than one character, it is not necessary that the gene shows dominance for all the characters.
Reason (R) :- Dominance is not an autonomous feature of a gene or the product that it has information for. It also depends on characters we choose to study.

(1) Assertion is true but reason is false

(2) Both A & R are false

(3) A & R both are true but R is not correct explanation of A

(4) A & R both are true and R is correct explanation of A

3) Which of the following statement is not always true for gene :

(1) Occur in pairs.

(2) Independent pairs segregate independently of each other.

(3) Pairs of gene segregate at the time of gamete formation.

(4) One pair segregates independently of another pair.

4) If a pure round, green seeded pea plant is crossed with pure wrinkled yellow seeded pea plant, their F_1 is then self - pollinated then in F_2 :
(a) 9/16 round, yellow plants produced.
(b) 4/16 double heterozygotes produced.
(c) Phenotypic ratio is deviated from 9 : 3 : 3 : 1.
(d) Only 1/16 wrinkled yellow seeded plant produced.
How many statements are incorrect ?

(1) a, b

(2) a, c

(3) c, d

(4) a, c, d

5) Match these columns -

(a)	Test cross	(i)	Tt × Tt	(p)	1 : 1 : 1 : 1
(b)	Monohybrid cross	(ii)	Rr × Rr	(q)	3 : 1
(c)	Dihybrid cross	(iii)	RrYy × rryy	(r)	1 : 2 : 1
(d)	Incomplete dominance	(iv)	RrYy × RrYy	(s)	9 : 3 : 3 : 1

(1) a-iii-p, b-i-q, c-iv-s, d-ii-r

(2) a-iii-p, b-ii-r, c-iv-s, d-i-q

(3) a-ii-r, b-i-q, c-iii-s, d-i-p

(4) a-i-p, b-ii-q, e-iii-r, d-iv-s

6) Identify incorrect statements -
(a) By RrYy plant, four types of gametes RY, Ry, rY, ry produced each with a frequency of 50%.
(b) Two alleles of a gene pair are located on homologous sites on homologous chromosomes.
(c) Generally, closely located genes on a chromosome, assorted independently.
(d) The alleles I^A and I^B produce same form of sugar while allele i doesn't produce any sugar.
(e) Characters segregate while formation of gametes.

(1) a, c
(2) a, b, c
(3) a, c, d
(4) b, c, e

7) How many of the following gametes are not possible in case of parent with genotype AaRrGG ?
ARG, Arg, aRG, ARr, arG, arg, ArG

(1) Four
(2) Three
(3) Two
(4) One

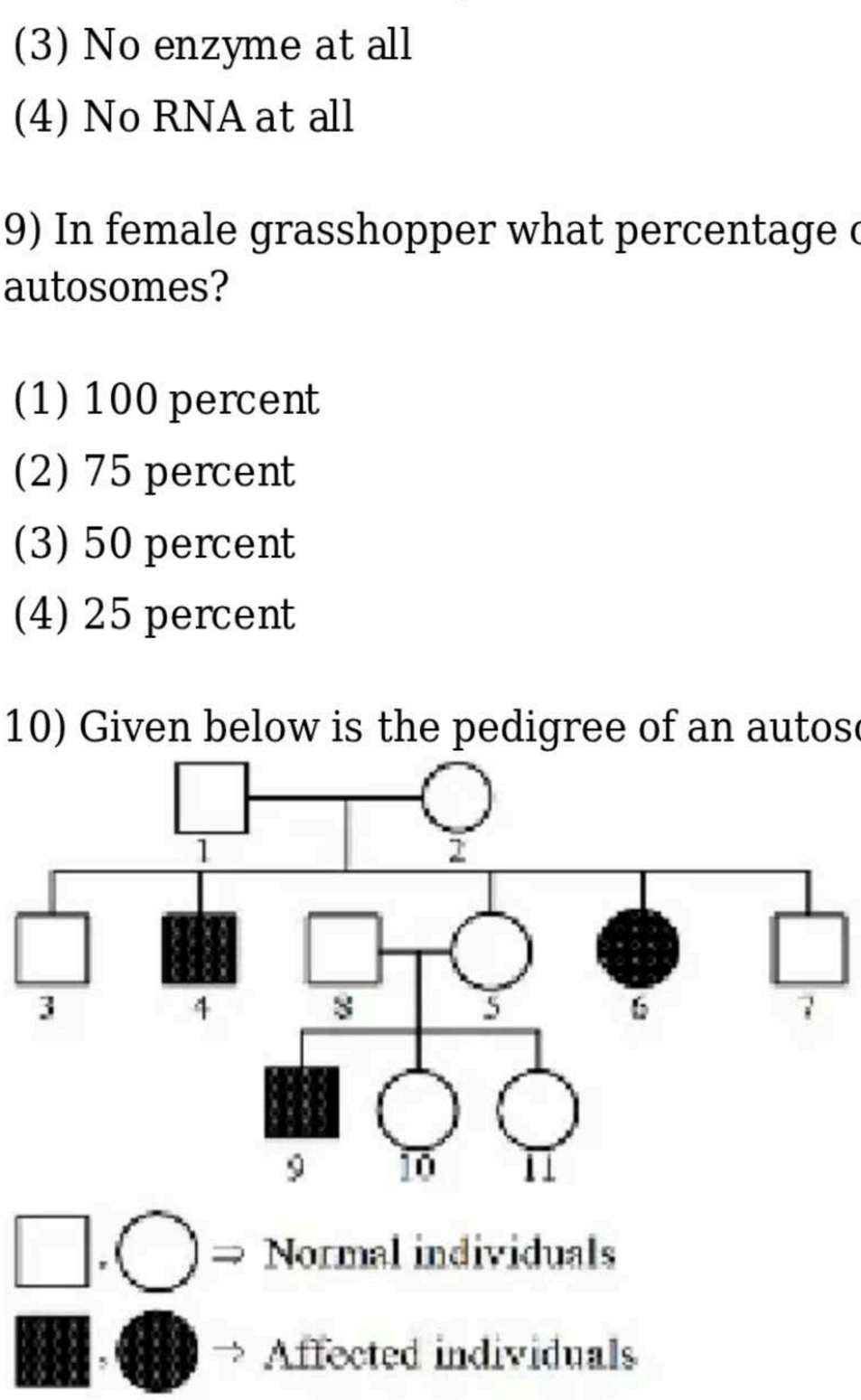
8) Modified allele is equivalent to unmodified allele if it produces -

(1) Normal enzyme
(2) Non-functional enzyme
(3) No enzyme at all
(4) No RNA at all

9) In female grasshopper what percentage of gametes have one X-chromosome besides the autosomes?

(1) 100 percent
(2) 75 percent
(3) 50 percent
(4) 25 percent

10) Given below is the pedigree of an autosomal recessive trait - sickle cell anaemia.



In the given pedigree the genotype of individuals 8 and 11 must be:

(1) 8 = AA, 11 = Aa

(2) 8 = AA, 11 = AA

(3) 8 = AA, 11 = AA or Aa

(4) 8 = Aa, 11 = Aa or AA

11) Among the progenies of couple the daughters are color blind but the sons are normal. What should be the genotype of this couple?

(1) Male - $X^{+}Y$ Female - $X^{+}X^{c}$

(2) Male - $X^{C}Y$ Female - $X^{c}X^{c}$

(3) Male - $X^{c}Y$ Female - $X^{+}X^{c}$

(4) Male - $X^{+}Y$ Female - $X^{+}X^{+}$

12) A yellow seeded and violet flowered pea plant was crossed with a green seeded and white flowered pea plant. The progeny showed plants with two different phenotypes - yellow seeds and violet flowers and green seeds with voilet flowers. The parent plant with yellow seeds and violet flowers must be :

(1) Double heterozygous

(2) Homozygous for seed colour and heterozygous for flower colour.

(3) Heterozygous for seed colour and homozygous for flower colour.

(4) Double homozygous

13) Given below are two statements: one is labelled as Assertion (A) and other is labelled as Reason (R):
Assertion (A): In honey bee the males have half the number of chromosomes than that of female.
Reason (R): In honey bee an unfertilised egg develops as a male by means of parthenogenesis.
In the light of the above statements, choose the correct answer from the option given below:

(1) Both (A) and (R) are true and (R) is the correct explanation of (A).

(2) Both (A) and (R) are true and (R) is not the correct explanation of (A).

(3) (A) is true but (R) is false.

(4) Both (A) and (R) are false.

14) Morgan hybridised yellow-bodied, white-eyed females Drosophila to brown-bodied, red-eyed males Drosophila and intercrossed their F_1 progeny. he observed that the two genes did not segregate independently of each other and the F_2 ratio deviated very significantly from the 9:3:3:1 ratio (expected when the two genes are independent).
Given below are two statements related to the Morgan's experiment given above:
Statement-I: In F_2 progeny the proportion of parental gene combinations were much higher than the non-parental type.
Statement-II: Morgan attributed that more parental combinations in F_2 progeny is due to more chances of crossing over between genes present on a chromosome.
In the light of about statements, choose the most appropriate answer from the options given below.

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

15) In Mendel's monohybrid cross the ratio between homozygous and heterozygous offsprings in F_2 generation was :-

(1) 1 : 1

(2) 3 : 1

(3) 1 : 2 : 1

(4) 1 : 2

16) In a true breeding variety :-

(1) Allelic pair of genes are identical.

(2) Allelic pair of ganes are dissimilar.

(3) Genes are not present in pair.

(4) More than two alleles of a gene are present in an individual.

17)

	Disorder	Pattern	Symptoms	Effect
1	Colour blindness	A	Failure to discriminate between red and green colour	Defect in either red or/and green cone cells of retina
2	Haemophilia	Sex-linked recessive	B	Defective form of blood clotting factor VIII/IX
3	C	Autosomal recessive	D	Abnormal haemoglobin in RBCs
4	Phenylketonuria	Autosomal recessive	E	Defective form of enzyme phenylalanine hydroxylase
5	Thalassemia (α/β)	Autosomal recessive	Leads to anaemia	F

(1)
(A) Sex linked recessive
(B) Mental retardation
(C) Delayed blood clotting
(D) Sickle cell anaemia.
(E) Synthesis of α & β globin chains absent or reduced.
(F) Sickle shaped RBC leads to anaemia.

(2)
(A) Sex linked recessive
(B) Normal blood clotting
(C) Sickle shaped RBC leads to anaemia.
(D) Sickle cell anaemia.
(E) Normal synthesis of α & β globin chain.
(F) Mental retardation

(3)
(A) Sex linked recessive
(B) Delayed blood clotting
(C) Sickle cell anaemia.
(D) Sickle shaped RBC leads to anaemia.
(E) Mental retardation
(F) Synthesis of α & β globin chains of haemoglobin absent or reduced.

(4)
(A) Sex linked recessive
(B) Synthesis of α & β globin chains of haemoglobin absent or reduced.
(C) Sickle cell anaemia.
(D) Sickle shaped RBC leads to anaemia.
(E) Mental retardation
(F) Delayed blood clotting

18) Select the correct statement -

(1) The sex determination is honey bee is based on the number of set of chromosomes an individual receives.

(2) In humans, 22 pairs of chromosome are different in both males and female.

(3) Female heterogamety present in humans.

(4) Grass hopper is an example of XX-XY type of sex determination

19) Which of the following statement is incorrect?

(1) Besides the involvement of multiple genes polygenic inheritance also takes into account the influence of environment.

(2) An example of pleiotropy is phenylketonuria.

(3) In pea plants heterozygotes (Bb) produced round seeds with large size of starch grain in seeds.

(4) In a large number of insect the mechanisms of sex determination is of the XO type.

20) Which of the following statements are correct about Down's syndrome ?
18. (Trisomy of 18).
(b) This disorder was first described by Langdon Down.
(c) The affected individual is short statured with large round head.
(d) Furrowed tongue and partially open month.

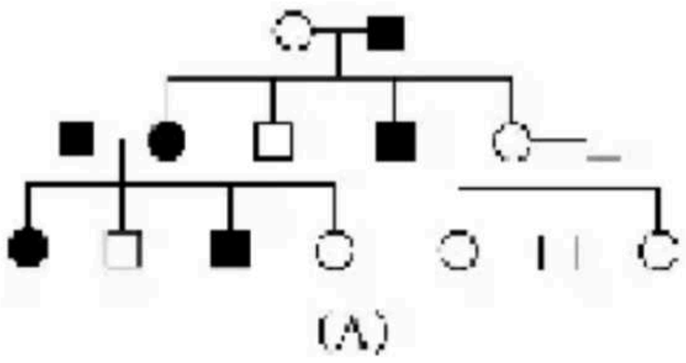
(1) All a, b, c, d are correct

(2) only a and b are correct

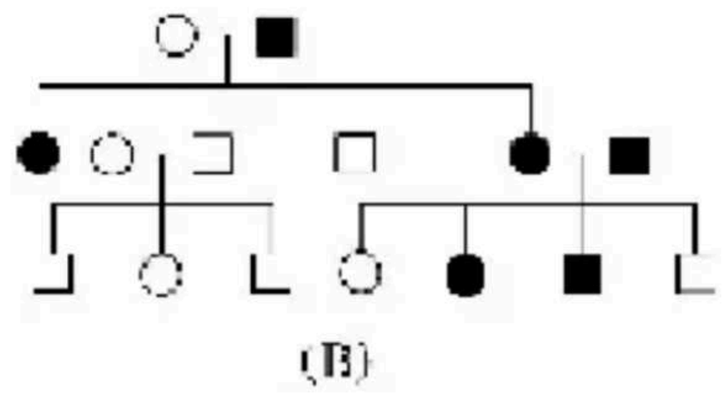
(3) only b and c are correct

(4) only b and d are correct

21) A and B are pedigree analysis of 2 diseases. Select the correct statement -



(A)



(B)

(1) A represented autosomal dominant trait while B represents autosomal recessive trait.

(2) A represents Y-linked trait while B represents X - linked recessive trait.

(3) A represents autosomal dominant trait white B represents X - linked dominant trait.

(4) A represents autosomal recessive trait while B represents autosomal dominant trait.

22) Consider following genetic disorders :
Haemophilia, Sickle cell Anaemia, Cystic fibrosis, Thalassemia, Phenylketonuria, Polydactyly, Klinefelter's syndrome, Turner's syndrome.
How many of the above disorder are Mendelian disorder ?

(1) 6

(2) 2

(3) 8

(4) 3

23) Fill in the blanks :
The VNTR belongs to a class of satellite DNA referred a as _____. Size of VNTR varies in size from _____.

(1) Micro satellite, 0.1 to 20 kb

(2) Mini satellite, 0.1 to 20 kb

(3) Micro satellite, 1 to 20 kb

(4) Mini satellite, 1 to 20 kb

24) Given below are two statements :
Statement-I :- Insertion or deletion of one or two bases changes the reading frame from the point of insertion or deletion.
Statement-II :- Sickle cell anemia is an example of frameshift mutation.

(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I & statement II are incorrect

(4) Both statement I & statement II are correct

25) Given below are two statements :

Statement-I :- Polymorphism in DNA sequence is the basis of genetic mapping of human genome as well as DNA finger printing.
Statement-II :- If a non heritable mutation is observed in a population at high frequency it is also referred as DNA polymorphism.

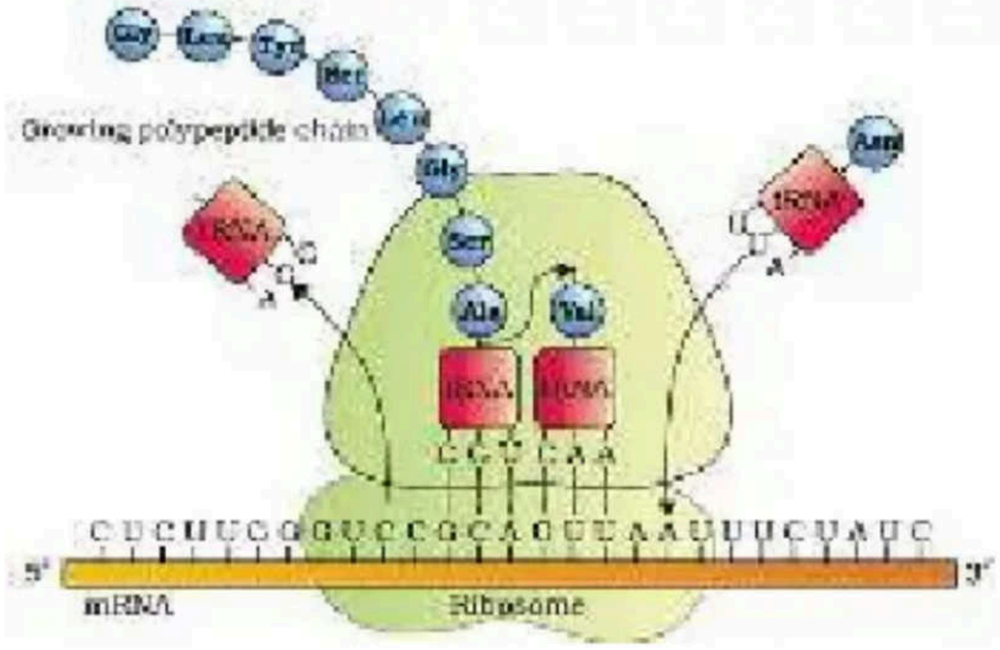
(1) Statement I is correct but statement II is incorrect

(2) Statement I is incorrect but statement II is correct

(3) Both statement I & statement II are incorrect

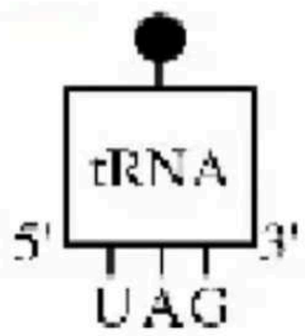
(4) Both statement I & statement II are correct

26)

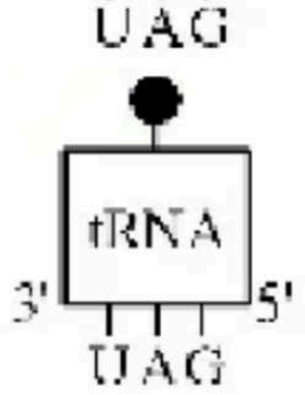


Which of the following t-RNA binds with last codon of the m-RNA :

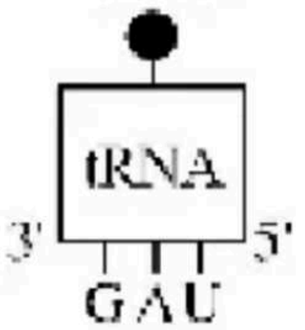
(1)



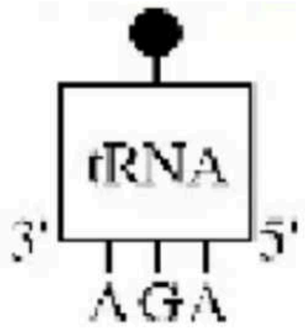
(2)

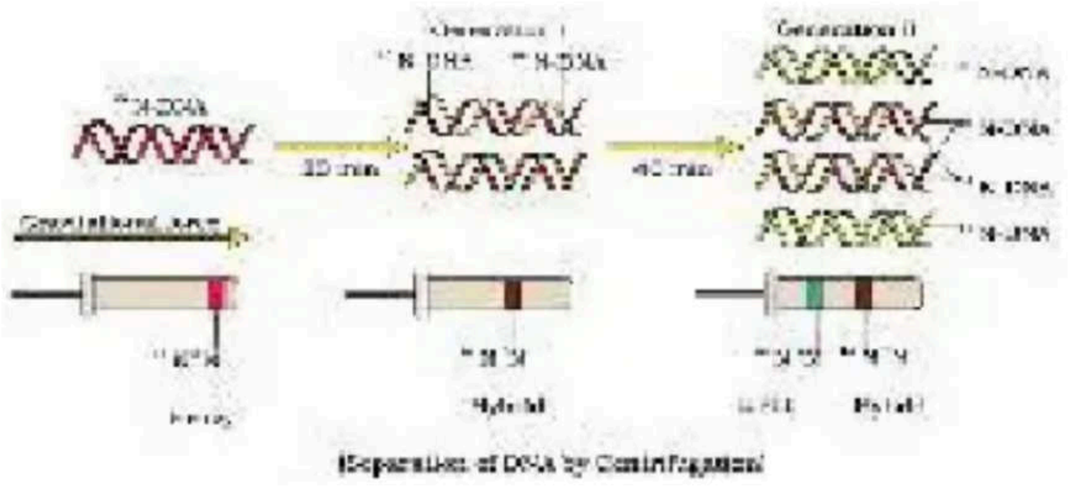


(3)



(4)



27) 

If the given experiment continues for 20 more minutes, then what ratio will be obtain between light, hybrid and heavy DNA ?

(1) 3 : 1 : 0

(2) 1 : 3 : 0

(3) 6 : 2 : 1

(4) 0 : 1 : 3

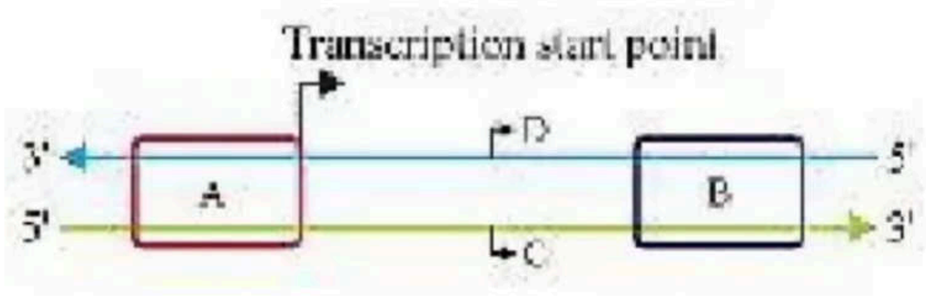
28) One of the hallmark of double Helix model of DNA was base pairing between the two strands of polynucleotide chains, However this proposition was also based on the observation of Erwin Chargaff for double stranded DNA.
Given below are two statements about this concept :
Statement-I :- In a dsDNA of 6 kb if we know the number of adenine then we can calculate the number of other N-bases.
Statement-II :- According to Chargaff rule we can easily understand the arrangement of N-base in single stranded DNA.
Choose the most appropriate answer from the option given below :

(1) Statement-I is correct but statement-II is incorrect

(2) Statement-I is incorrect but statement-II is correct

(3) Both statement-I & statement-II are incorrect

(4) Both statement-I & statement-II are correct

29) 

In above diagram which of the following point represent coding strand :

(1) A

(2) B

(3) C

(4) D

30) **Statement-1 :** By switching the position of promoter with terminator, the definition of coding and template strands could be reversed.
Statement-2 : The structural gene in a transcription unit could be said as monocistronic (mostly in eukaryotes) or polycistronic (mostly in prokaryotes).

(1) Both statement-1 and statement-2 are correct.

(2) Both statement-1 and statement-2 are incorrect.

(3) Statement-1 is correct but statement-2 is incorrect.

(4) Statement-1 is incorrect but statement-2 is correct.

31) Find out the correct match from the column-I, II and III.

	Column-I		**Column-II**		**Column-III**
I.	Meselson and Stahl	(a)	Bacteriophages	(i)	Proved that DNA replication is semiconservative in bacteria
II.	Hershey and Chase	(b)	*Vicia faba*	(ii)	Proved that DNA in chromosome also replicate semi conservatively
III.	Griffith	(c)	*E.coli.*	(iii)	Transformation principle
IV.	Taylor	(d)	*Streptococcus pneumoniae*	(iv)	Proof that DNA is genetic material

(1) I-c-ii, II-a-iii, III-d-iv, IV-b-i

(2) I-a-i, II-b-iii, III-d-iv, IV-c-i

(3) I-c-i, II-a-iv, III-d-iii, IV-b-ii

(4) I-b-iv, II-c-ii, III-a-i, IV-d-iii

32) Match the column -

	Column-I		**Column-II**
(a)	Euchromatin	(i)	Loosely packed
(b)	Heterochromatin	(ii)	Lightly stained
		(iii)	Transcriptionally inactive
		(iv)	Transcriptionally active
		(v)	Densely packed
		(vi)	Darkly stained

(1) (a)–(ii),(v),(iii) (b)–(i),(vi),(iii)

(2) (a)–(i),(ii),(iii) (b)–(iv),(v),(vi)

(3) (a)–(i),(ii),(iv) (b)–(iii),(v),(vi)

(4) (a)–(vi),(i),(iii) (b)–(ii),(v),(iii)

33) During Meselson and Stahl experiments, heavy DNA was distinguished from normal DNA by centrifugation in :

(1) Cs OH gradient

(2) Cs Cl gradient

(3) $^{14}NH_4Cl$ gradient

(4) $^{15}NH_4Cl$ gradient

34) Which of the following is not correct ?

(1) $\frac{A}{T} = 1$

(2) A + T = G + C

(3) A + G = C + T

(4) $\frac{G}{C} = 1$

35) If % of cytosine in a dsDNA is 18%, then % of adenine will be :

(1) 32%

(2) 64%

(3) 36%

(4) 23%

36) Which of the following is NOT true about prokaryotic transcription?

(1) Only one type of RNA polymerase synthesizes all types of RNA.

(2) Sigma factor is required only for initiation, not elongation.

(3) Transcription and translation can occur simultaneously.

(4) RNA polymerase requires primers to initiate transcription.

37) Why are both strands of DNA not copied during transcription?

(1) Because both strands are identical and would produce the same RNA.

(2) information transfer.

(3) Because RNA polymerase can bind only to one strand at a time.

(4) Because DNA strands are too short for transcription on both strands.

38) Which of the following provide energy for polymerisation reaction in transcription ?

(1) Deoxyribonucleotides triphosphate

(2) Deoxyribonucleotides monophosphate

(3) Ribonucleotides monophosphate

(4) Ribonucleotides triphosphate

39) Which of the following statement is not correct ?

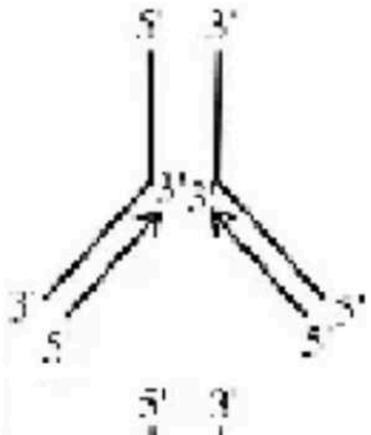
(1) If an inheritable mutation is observed in a population at high frequency, it is referred to as DNA polymorphism.

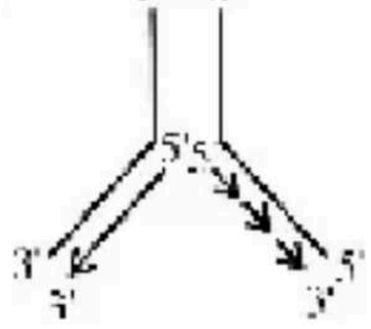
(2) DNA polymorphisms play no role for evolution & speciation.

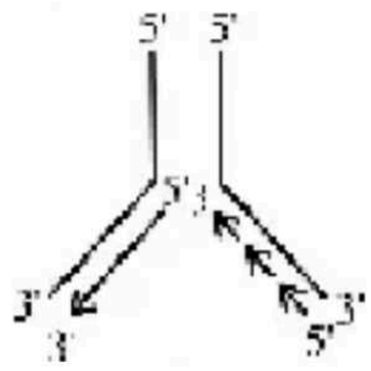
(3) The technique of DNA finger printing was initially developed by Alec Jeffrey.

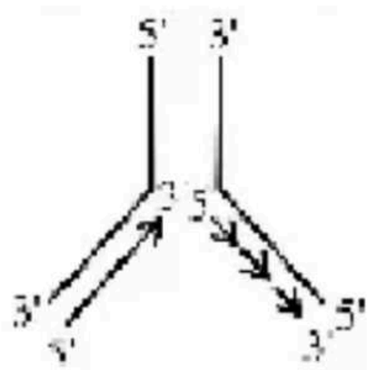
(4) The VNTR belongs to a class of satellite DNA referred to as mini-satellite.

40) Which of the following diagram correctly explain process of DNA replication :

(1) 

(2) 

(3) 

(4) 

41) In *E.coli*, if a mutation occurs in the ligase synthesizing gene, then which of the following effects will be seen in that bacterial cell ?

(1) Origin of replication is affected

(2) Gene expression is affected

(3) Discontinuous strand is not synthesized

(4) The discontineously synthesised fragments are not converted into continuous fragment

42) An *E.coli* with hybrid DNA (N_{14}/N_{15}) is transfered into $^{14}NH_4Cl$ medium. After 60 minutes how many *E.coli* cells will be present that contain hybrid DNA ?

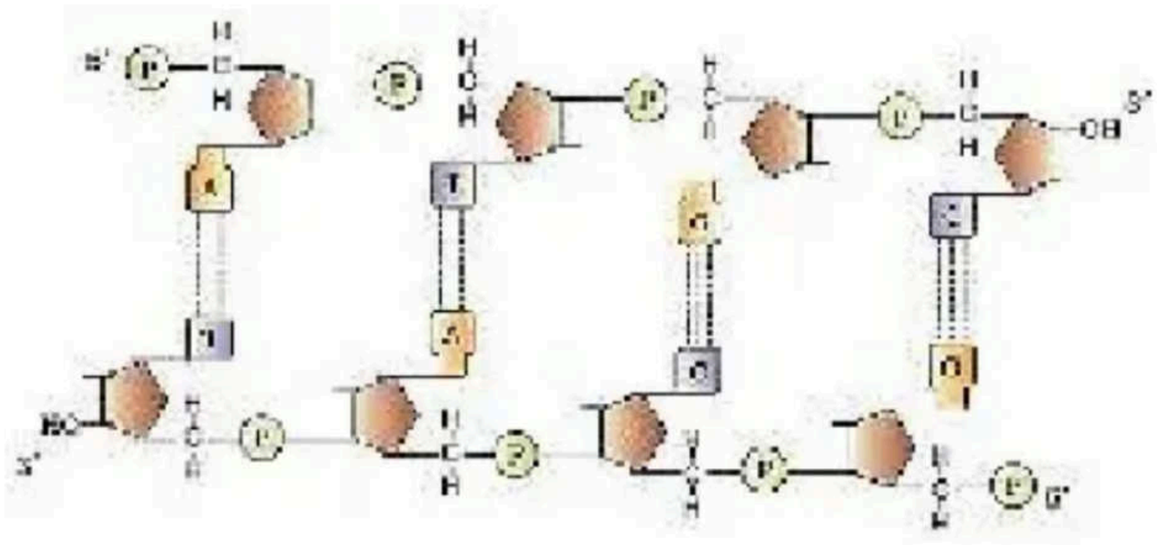
(1) 2

(2) 6

(3) 8

(4) 1

43)



From this dsDNA molecule if we remove one A = T base pair then how many hydrogen bonds and phosphodiester bonds remain in it :

(1) 10, 6

(2) 8, 6

(3) 8, 4

(4) 4, 8

44) 5'AUGAGGGUGUGAUUGUUUUAA3'

If given m-RNA going through translation process then how many amino acid long polypeptide chain will be synthesize :

(1) 6

(2) 4

(3) 5

(4) 3

45) During the study of gene expression in a cell, a researcher observed that the complete RNA was not translated into amino acid and some untranslated part occured in the m-RNA at more than one place. Which of the following option is true about this observation :

(1) Untranslated part may be Exon.

(2) The cell may be prokaryotic with split gene.

(3) The cell may be Eukaryotic with split gene but had not undergone splicing process.

(4) This phenomenon is not possible.

46)

During isolation, precipitated DNA can be separated out by a process called _____.

Fill in the blank with suitable option :

(1) Elution

(2) Spooling

(3) Electrophoresis

(4) Biolistics

47) A piece of DNA if needed to be propagated during recombinant DNA procedures requires a vector, because vector provide :-

(1) Origin of replication site

(2) Origin of transcription site

(3) Origin of translation site

(4) Provides stability

48) Which of the following is not a property of good cloning vector DNA :-

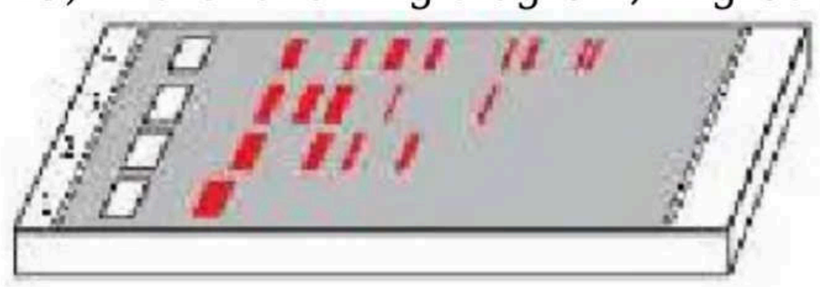
(1) Origin of replication (ori)

(2) It contain a selectable marker

(3) Single recognition site for commonly used restriction enzyme

(4) Many recognition sites for commonly used restriction enzyme

49) In the following diagram, migration of undigested DNA is seen in -



(1) Lane 4

(2) Lane 3

(3) Lane 2

(4) Lane 1

50) Which compound is used to stain DNA in Agarose Gel ?

(1) Ethidium bromide

(2) $NaNO_3$

(3) Ammonium nitrate

(4) Polyethylene glycol

51) The first restriction endonuclease reported was :

(1) Hind II

(2) EcoRI

(3) Hind III

(4) BamHI

52) **Assertion (A) :** *Agrobacterium tumifaciens* infects all broad leaved agriculture crops such as tomato, soyabean, sunflower, cotton and cereals.
Reason (R) : The Ti-plasmid of *agrobacterium tumifaciens* is responsible for inducing crown gall disease in plants.

(1) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(2) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(3) (A) is correct but (R) is incorrect.

(4) (A) is incorrect but (R) is correct.

53) Which of the following is true :-

(1) At present about 20 recombinant therapeutics have been approved for human use the world over.

(2) In India 5 recombinant therapeutic are presently being marketed

(3) Genetically engineered insulin consists three short polypetide chains

(4) C peptide is not present in mature insulin

54) **Statement - I :** Selectable markers help in identifying and eliminating non-transformants.
Statement - II : Antibiotic resistance genes in vectors act as selectable markers.

(1) Both **Statement I** and **Statement II** are incorrect.

(2) **Statement I** is correct but **Statement II** is incorrect.

(3) **Statement I** is incorrect but **Statement II** is correct.

(4) Both **Statement I** and **Statement II** are correct.

55) Match the column and choose correct option –

	Column-I		**Column-II**
(A)	Microinjection	(i)	Way to introduce DNA into host cell
(B)	PCR	(ii)	Separation of DNA
(C)	Downstream processing	(iii)	Amplification of gene
(D)	Gel electrophoresis	(iv)	Separation & purification of product

(1) (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)

(2) (A)-(ii), (B)-(iii), (C)-(in), (D)-(i)

(3) (A)-(i), (B)-(iii), (C)-(iv), (D)-(ii)

(4) (A)-(iv), (B)-(i), (C)-(ii), (D)-(iii)

56) Match the column A from column B

	Column-A		**Column-B**
(I)	ELISA	(a)	Nematode resistance plant
(II)	RNA-i	(b)	Cotton bollworm
(III)	cryIAc	(c)	Molecular diagnosis
(IV)	α-1 antitrypsin	(d)	To treat emphysema

(1) I → c, II → a, III → b, IV → d

(2) I → c, II → b, III → d, IV → a

(3) I → a, II → b, III → c, IV → d

(4) I → d, II → c, III → a, IV → b

57) When a desired DNA is inserted within the coding sequence of an enzyme, β-galactosidase:-
I. This result into inactivation of the β-gal gene
II. This is called insertional inactivation

III. In the presence of insertion, the colonies do not produce any colour

(1) I only

(2) I and II

(3) I and III

(4) I, II and III

58) Today, we know more than ___B___ restriction enzymes that have been isolated from over ___A___ strains of bacteria.

(1) A-900, B-230

(2) A-1500, B-500

(3) A-230, B-900

(4) A-500, B-1500

59) Due to insertion of desired gene within selectable marker gene of vector :-

(1) Selectable marker gene become inactive or lose their functions

(2) Selectable marker gene become active or perform their function

(3) This is called insertional inactivation

(4) (1) and (3) both

60) Select the correct match option :-

(1) Gene therapy → Separation of DNA fragments

(2) RNA interference → Cellular defence

(3) ELISA → Diagnostic test for genetic disorder

(4) Gel electrophoresis → Allows correction of a gene defect.

61) How many of the following enzymes are not used for isolation of DNA from E.coli?

(A) Cellulase (B) Lysozyme
(C) Chitinase (D) Taq DNA polymerase
(E) E.CORI (F) DNA ligase
(G) Protease (H) Ribonulease

(1) Four

(2) Three

(3) Six

(4) Five

62) Read the given below statements:

(i) Biolistics method of gene transfer is suitable for animal cell.
(ii) Microinjection is a method of gene transfer into the nucleus of an animal cell.
(iii) Gene gun method is suitable for plant cell
(iv) Microparticles of gold or tungsten are used in biolistics method of gene transfer.
(v) In order to force bacteria to take up the plasmid the bacterial cell must first be made 'competent'

to take up DNA.
How many above statement are correct ?

(1) Two

(2) Three

(3) Four

(4) Five

63) Which of the following statements are correct ?
(a) Agrobacterium-mediated gene transfer involves the use of a bacterium to transfer genes into dicot plant cells.
(b) DNA polymerase derived from *Thermus aquaticus* is used in PCR due to its heat stability.
(c) Bioreactors are used for large scale cultivation of micro-organism for biotechnological processes.
(d) 2,00,000 varieties of basmati rice are grown in India.
(e) The sieving effect during gel electrophoresis is due to ethidium bromide.

(1) a, b, e

(2) b, c, d

(3) a, b, c

(4) d and e only

64) Which of the following genetically engineered organisms was used by Eli Lilly to prepare human insulin ?

(1) Bacterium

(2) Yeast

(3) Virus

(4) Phage

65) In order to obtain virus-free plants through tissue culture the best method is :-

(1) Protoplast culture

(2) Embryo rescue

(3) Anther culture

(4) Meristem culture

66) The use of bioresources by multinational companies and other organization without proper authorisation from countries and people concered with compensatory payment it is known as :-

(1) Biopiracy

(2) Biopatent

(3) Biowar

(4) Bioremediation

67) The proteins encoded by the genes cry I Ac and cry II Ab are effective against :-

(1) Corn borer

(2) Cotton bollworms

(3) Nematodes

(4) Mosquitoes

68) The two polypeptides of human insulin are linked together by :-

(1) Hydrogen bonds

(2) Phosphodiester bond

(3) Glycosidic bond

(4) Disulphide bridges

69) The Indian parliament has recently cleared the ____ amendment of the Indian patents Bill that takes such issues into consideration, including patent terms emergency provisions and research and development initiative :-

(1) First

(2) Third

(3) Second

(4) Fifth

70) **Assertion (A):** Transgenic animals are made that carry genes which make them more sensitive to toxic substances than non-transgenic animals.
Reason (R): Toxicity testing in such animals will allow us to obtain results in less time
In the light of the above statements, choose the correct answer from the options given below:

(1) Both A and R are true and R is the correct explanation of A

(2) Both A and R are true but R is NOT the correct explanation of A

(3) A is true but R is false

(4) A is false but R is true

71) Select the **incorrect** statement regarding gene therapy.

(1) The first clinical gene therapy was given in 1990 to a 4-year old girl with adenosine deaminase (ADA) deficiency.

(2) ADA deficiency can be permanetly cured by infusion of genetically engineered Lymphocytes into patients.

(3) Gene therapy is a collection of methods that allows correction of a gene defect that has been diagnosed in a child/embryo.

(4) In ADA deficinecy gene therapy retroviruses are used as vectors.

72) Match the column I and Column II :-

	Column-I		**Column-II**
(A)	α-1-autitrypsin	(I)	Digestion of plant cell wall
(B)	Transgenic mice	(II)	Treat emphysema
(C)	Lysozyme	(III)	Testing the safety of vaccines

(D)	Cellulase	(IV)	Digestion of bacterial cell wall

(1) A-(II), B-(III), C-(IV), D-(I)

(2) A-(III), B-(II), C-(I), D-(IV)

(3) A-(II), B-(III), C-(I), D-(IV)

(4) A-(III), B-(II), C-(IV), D-(I)

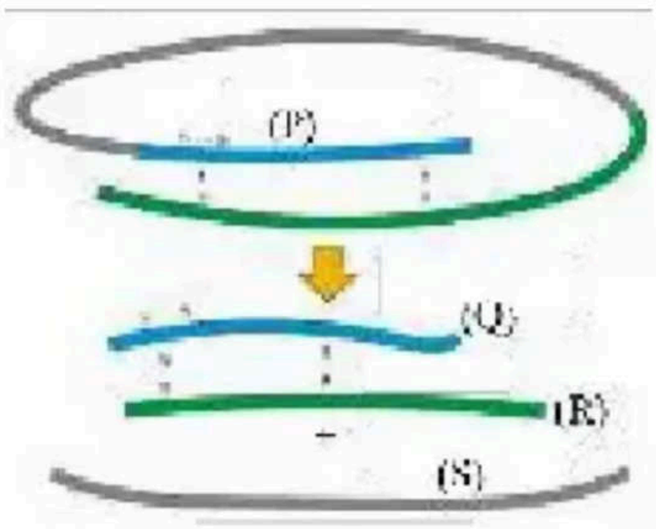
73) Two statements are given below:
Statement I: Silencing of a specific mRNA due to a complementary ssRNA molecule that binds to and prevents translation of the mRNA (silencing).
Statement II: The source of this complementary RNA could be from an infection by viruses having RNA genomes or mobile genetic elements (transposons) that replicate via an RNA intermediate.
In the light of the above statements, choose the most appropriate answer from the options given below:

(1) Both statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct



74) Read the following statements about the given diagram.
(i) P-proinsulin secreted by pancreases, S-not form when insulin is produced by recombinant DNA technology
(ii) Q-Peptide chain is not present in the mature insulin
(iii) Diagram represent the maturation of pro-insulin into insulin
(iv) Insulin is made up of 51 amino acids
How many statements are correct ?

(1) 1

(2) 2

(3) 3

(4) 4

75) How many in the given list below are benefits of doing genetic modification in plants ?
List : (A) Nutritional enhancement of food.
(B) Reduction in post harvest loss
(C) Pest resistance
(D) Tolerance to drought
(E) Increase in mineral usage efficiency
Choose the correct answer from the options given below.

(1) Five

(2) Four

(3) Two

(4) Three

76) Transgenic models exist for which for the following human diseases-
(a) AIDS (b) Cystic fibrosis
(c) Rheumatoid arthritis (d) Alzheimer's
(e) Emphysema

(1) a, b, and c

(2) b, c and d

(3) b, c, d and e

(4) a, b, d and e

77) Penicillin is obtained from :-

(1) Yeast

(2) Bacteria

(3) Fungi

(4) Algae

78) Which of the following is also called brewer's yeast?

(1) *Penicillum notatum*

(2) *Acetobactoer aceti*

(3) *Saccharomyces cerevisiae*

(4) *Trichoderma*

79) **Assertion (A) :** Statins produce by the yeast have been commercialised as blood-cholesterol lowering agents.
Reason (R) : Statins competitively inhibit the enzyme responsible for synthesis of cholesterol.
Select the correct option :-

(1) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

(2) (A) is correct but (R) is incorrect.

(3) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(4) Both (A) and (R) are incorrect.

80) **Assertion(A):** Microbes, especially yeasts have been used from time immemorial for production of beverages like wine, beer, whisky, brandy or rum.
Reason(R): Wine and beer are produced without distillation whereas whisky, brandy and rum are produced by distillation of the fermented broth.

(1) Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.

(2) Both Assertion and Reason are true and Reason is the correct explanation of Assertion.

(3) Assertion is true but Reason is false

(4) Assertion is false but Reason is true

81) The bottle juices are clarified by the use of :-

(1) Lipase

(2) Proteases

(3) Pectinases

(4) Both 2 and 3

82) Which of the following statements is **correct** ?

(1) The full potential of penicillin as an effective antibiotic was established by Alexander Fleming during world war-II.

(2) Different types of alchoholic drinks can be obtained depending on the type of the raw material used for fermentation and the type of processing.

(3) Statins competitively inhibit the enzyme responsible for immune response.

(4) *Aspergillus niger* is used in the commercial production of aspartic acid.

83) How many of the following is/are correct regarding BOD?
a. BOD is the measure of the organic matter present in the water
b. Greater the BOD of waste water, more is its polluting potential.
c. Refers to the amount of oxygen that would be consumed if all the inorganic matter in one liter of water were oxidised by bacteria.

(1) only a

(2) only b

(3) a and b

(4) b and c

84) Match the following :-

	Column-I		**Column-II**
(a)	Citric acid	(i)	Trichoderma polysporum
(b)	Butyric acid	(ii)	Aspergillus niger
(c)	Cyclosporin-A	(iii)	Monascus purpureus
(d)	Statins	(iv)	Clostridium butylium

(1) a-iv, b-ii, c-i, d-iii

(2) a-ii, b-iv, c-i, d-iii

(3) a-ii, b-iv, c-iii, d-i

(4) a-iii, b-iv, c-i, d-ii

85) Given below are the two statements :-
Statement-I : The majority of baculoviruses used as biocontrol agent are in the genus nucleopolyhedrovirus.

Statement-II : The use of baculoviruses is desirable when an ecologically sensitive area is being treated.
Select the correct option :-

(1) Both statement-I and statement-II are incorrect.

(2) Both statement-I and statement-II are correct.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

86) **Statement-I:** In secondary sewage treatment vigorous aeration is done to promote the growth of aerobic microbes.
Statement-II: Aerobic microbes helps in the formation of flocs that digest organic matter in the effluent.

(1) Both statement-I and statement-II are correct.

(2) Both statement-I and statement-II are incorrect.

(3) Statement-I is correct but statement-II is incorrect.

(4) Statement-I is incorrect but statement-II is correct.

87) Which of the following organisms can fix nitrogen?
A. Azobacter
B. Glomus
C. Nostoc
D. Anabaena
E. Trichoderma
Choose the correct answer from the options given below:

(1) A and B only

(2) A, B, C, D and E only

(3) A,C and D only

(4) B and E only

88) Which of the following statement(s) are correct?
A. The 'Roquefort Cheese' are ripened by growing a specific fungi on them.
B. The puffed-up-appearance of dough is due to the production of alcohol.
C. Wine and beer are produced without distillation.
D. Statins produced by the yeast monascus purpureus.
E. Biogas is a mixture of gases containing predominantly carbon dioxide.
Choose the correct answer from the options given below.

(1) A, C, D only

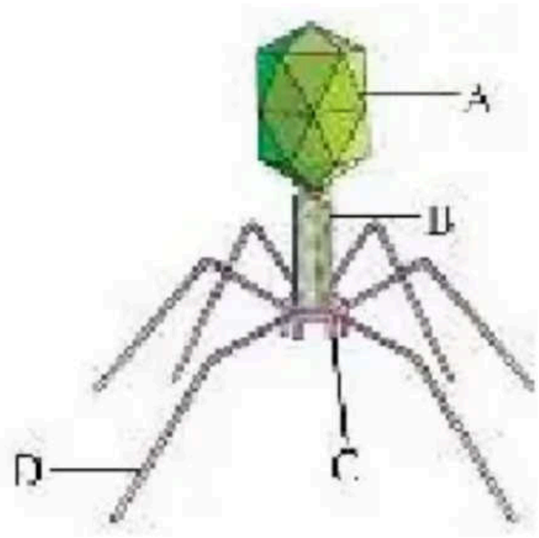
(2) B, C, D only

(3) C, E, A only

(4) D, E, B only

89) Given diagram is representation of a bacteriophage. Which one of following option is correct

about naming of it's parts.



(1) A-Collar, B-Head, C-Prongs, D-Pins

(2) A-Head, B-Collar, C-Pins, D-Prongs

(3) A-Pins, B-Prongs, C-Head, D-Collar

(4) A-Pins, B-Head, C-Prongs, D-Collar

90) Recombinant DNA technology involves several steps in specific sequence. What is the correct sequence for r-DNA technology ?
a → Down stream processing.
b → Cutting of DNA at specific location.
c → Obtaining the foreign gene product.
d → Isolation of the genetic material.
e → Insertion of r-DNA into the host cell.

(1) b → d → e → a → c

(2) d → b → e → c → a

(3) d → e → d → a → c

(4) e → d → a → c → b

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	**2**	**4**	**4**	**2**	**1**	**3**	**3**	**2**	**2**	**1**	**3**	**1**	**3**	**2**	**3**	**1**	**3**	**4**	**3**	**2**
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	**4**	**1**	**1**	**2**	**4**	**3**	**3**	**1**	**2**	**2**	**3**	**1**	**1**	**1**	**4**	**1**	**3**	**1**	**4**	**1**
Q.	41	42	43	44	45															
A.	**1**	**2**	**1**	**3**	**1**															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	**1**	**2**	**4**	**1**	**3**	**2**	**4**	**1**	**4**	**4**	**4**	**3**	**2**	**3**	**3**	**2**	**2**	**1**	**3**	**4**
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	**4**	**3**	**1**	**2**	**1**	**2**	**3**	**4**	**2**	**4**	**1**	**3**	**1**	**3**	**3**	**4**	**3**	**2**	**1**	**4**
Q.	86	87	88	89	90															
A.	**2**	**2**	**4**	**2**	**1**															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	**1**	**4**	**4**	**3**	**1**	**3**	**2**	**1**	**1**	**4**	**3**	**3**	**1**	**3**	**1**	**1**	**3**	**1**	**3**	**4**
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	**3**	**1**	**2**	**1**	**1**	**2**	**1**	**1**	**3**	**1**	**3**	**3**	**2**	**2**	**1**	**4**	**2**	**4**	**2**	**4**
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	**4**	**4**	**3**	**4**	**3**	**2**	**1**	**4**	**4**	**1**	**1**	**4**	**4**	**4**	**3**	**1**	**4**	**3**	**4**	**2**
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	**4**	**3**	**3**	**1**	**4**	**1**	**2**	**4**	**3**	**1**	**2**	**1**	**4**	**3**	**1**	**2**	**3**	**3**	**3**	**1**
Q.	171	172	173	174	175	176	177	178	179	180										
A.	**4**	**2**	**3**	**2**	**2**	**1**	**3**	**1**	**2**	**2**										

SOLUTIONS

PHYSICS

1) $\oint \vec{E}.\overrightarrow{d\ell} = -\frac{d\phi}{dt}$

2)

The velocity of e/m wave is equal to vel. of light in vaccume. In medium it change.

3) As $\vec{E} \perp \vec{B}$ so $\vec{E}.\vec{B} = 0$

So direction of $\vec{E} \Rightarrow \frac{\hat{i}}{2} - \frac{\sqrt{3}\hat{j}}{2}$

and $C = \frac{E_o}{30}$

4)

Radiowaves → Cellular phones
Microwaves → Aircraft navigation
Infraredwaves → LED
Ultraviolet waves → Water purifiers

5) All EM Waves travels with the same speed.

6) $\vec{E}_1 = E_0\hat{j}\cos(\omega t - kx)$

Its corresponding magnetic field will be

$\vec{B}_1 = \frac{E_0}{c}\hat{k}\cos(\omega t - kx)$

$\vec{E}_2 = E_0\hat{k}\cos(\omega t - ky)$

$\vec{B}_2 = \frac{E_0}{c}\hat{i}\cos(\omega t - ky)$

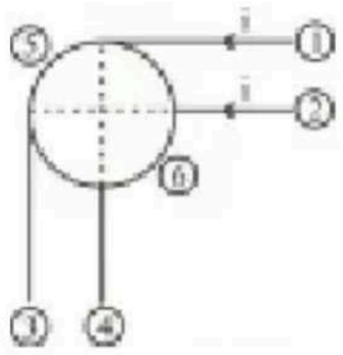
Net force on charge particle

$= q\vec{E}_1 + q\vec{E}_2 + q\vec{v} \times B_1 + q\vec{v} \times \vec{B}_2$

$= qE_0\hat{j} + qE_0\hat{k} + q(0.8c\hat{j}) \times \left(\frac{E_0}{c}\hat{k}\right) + q(0.8c\hat{j}) \times \left(\frac{E_0}{c}\hat{i}\right)$

$= qE_0\hat{j} + qE_0\hat{k} + 0.8qE_0\hat{i} - 0.8qE_0\hat{k}$

$\vec{F} = qE_0[0.8\hat{i} + 1\hat{j} + 0.2\hat{k}]$



7)

$B_2 = B_4 = 0$

$$\vec{B}_1 = \vec{B}_3 = \frac{\mu_0 i}{4\pi R}\odot$$

$$\vec{B}_5 = \frac{\mu_0 i}{8R}\odot$$

$$\vec{B}_6 = \frac{\mu_0 i}{8R}\otimes$$

$$B = B_1 + B_2 + B_3 + B_4 + B_5 + B_6$$

$$= \frac{\mu_0 i}{2\pi R}\odot$$

8) $$B = \frac{\mu_0 i}{2r} = \frac{\mu_0 \left(\frac{e}{T}\right)}{2r} = \frac{\mu_0 \left(\frac{eV}{2\pi r}\right)}{2r}$$

$$\Rightarrow B = \frac{\mu_0 eV \times mr}{4\pi r^2 \times mr}$$

$$B = \frac{\mu_0 e(mvr)}{4\pi m r^3}$$

$$B = \frac{\mu_0 eL}{4\pi m r^3}$$

9) For permanent magnet we prefer a material with high retentivity (so as to make a stronger magnet) and high coercivity (so that magnetization any not be wiped out easily). for electromagnet we prefer high saturated magnetism low coercivity and least possible area of hysteresis loop so that electromagnet develops high magnetization, is easily demagnetized and energy loss in a magnetization cycle is least. Therefore, P is suitable for making permanent magnet and Q for making electromagnet.

10)

Magnetic field at distance r from a long straight wire

$$B = \frac{\mu_0 I}{2\pi r}$$

Magnetic force on a moving charge

$F = qvB$ (since $v \perp B$)

Given:

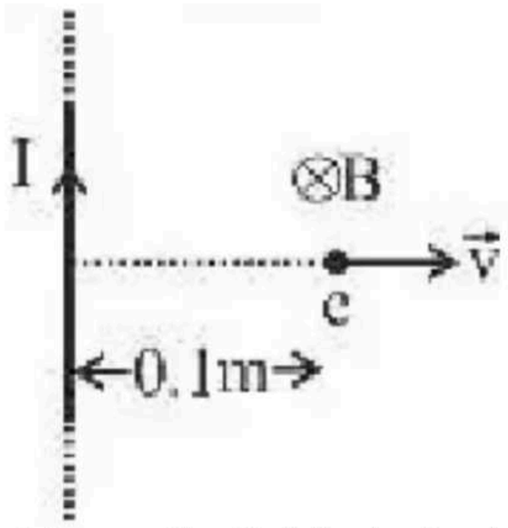
$I = 10$ A

$r = 0.1$ m

$v = 5.0 \times 10^6$ m s $^{-1}$

$q = 1.6 \times 10^{-19}$ C

$\mu_0 = 4\pi \times 10^{-7}$ T m A^{-1}



Magnetic field at electron position:

$$B = \frac{4\pi \times 10^{-7} \times 10}{2\pi \times 0.1}$$

$$B = \frac{40\pi \times 10^{-7}}{0.2\pi}$$

$B = 2 \times 10^{-5}$ T

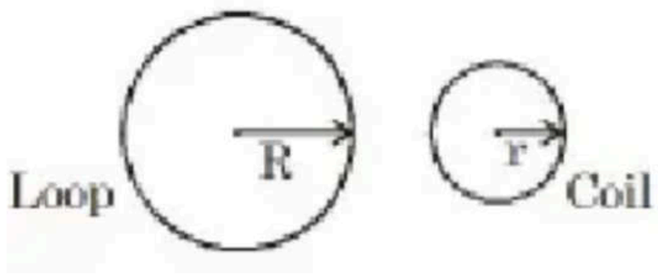
Force on electron:

$F = qvB$

$F = (1.6 \times 10^{-19})(5.0 \times 10^{6})(2 \times 10^{-5})$

$F = 1.6 \times 10^{-17}$ N

11)



$L = 2\pi R \quad L = N \times 2\pi r$

$R = Nr$

$$B_L = \frac{\mu_0 i}{2R} \qquad B_C = \frac{\mu_0 N i}{2r}$$

$$B_C = \frac{\mu_0 N^2 i}{2R}$$

$$\frac{B_L}{B_C} = \frac{1}{N^2}$$

12) $KE = q\Delta V$

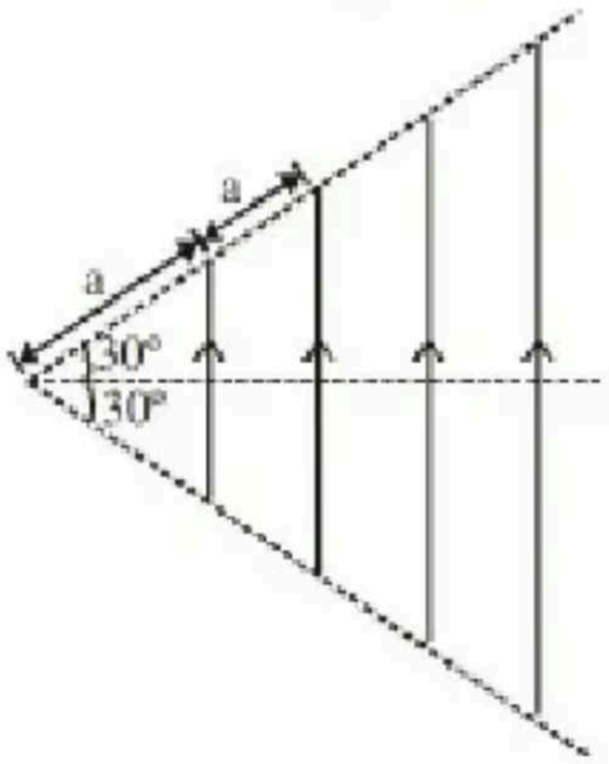
$$r = \frac{\sqrt{2mq\Delta V}}{qB}$$

$$r \propto \sqrt{\frac{m}{q}}$$

$$\frac{r_p}{r_\alpha} = \frac{1}{\sqrt{2}}$$

13) $$B_P = \left(\frac{\mu_0 i}{4r} + \frac{\mu_0 i}{4\pi r}\right) = \frac{\mu_0 i}{2r}\left(\frac{1}{2} + \frac{1}{2\pi}\right)$$

14)

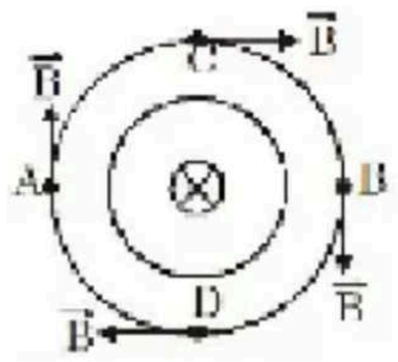


$$\vec{B} = \frac{\mu_0 i}{4\pi}\left(\frac{2\sin 30}{\cos 30}\right)\left(\frac{1}{a} - \frac{1}{2a} + \frac{1}{3a}\right)\hat{k}$$

$$= \frac{\mu_0 i}{4\pi a} \times \left(2 \times \frac{1}{\sqrt{3}}\right)(\ln 2)(\hat{k})$$

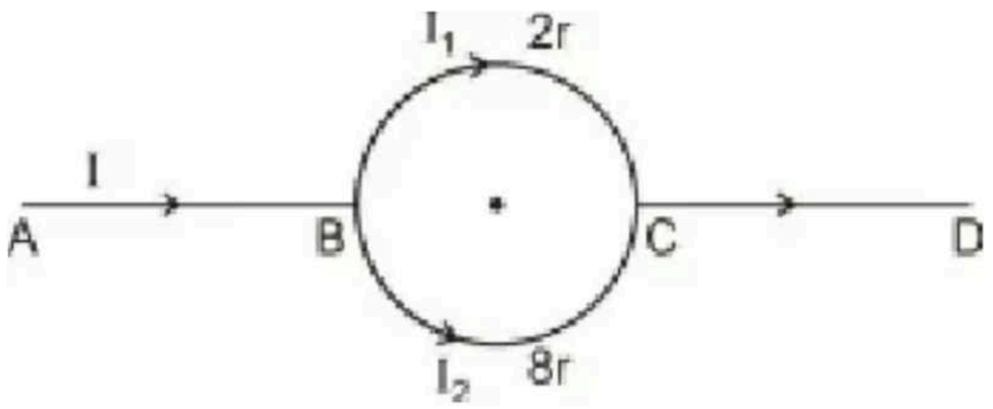
$$= \frac{\mu_0 i}{4\pi a(\sqrt{3})}(\ln 4)(\hat{k})$$

15)



Direction of magnetic field is clockwise by using right hand thumb rule.

16)



magnetic field due to wire AB and CD is zero at center.

$$I_1 = \frac{8r}{2r + 8r} \times I$$

$$\Rightarrow I_1 = \frac{8r}{10r} I \Rightarrow \frac{4}{5} I$$

$$I_2 = \frac{2r}{10r} I \Rightarrow \frac{I}{5}$$

$$B_1 = \frac{\mu_0\left(\frac{4I}{5}\right)}{4R} \Rightarrow \frac{\mu_0 I}{5R}\otimes$$

$$B_2 = \frac{\mu_0\left(\frac{I}{5}\right)}{4R} \Rightarrow \frac{\mu_0 I}{20R}\odot$$

$$B_{net} = \frac{\mu_0 I}{5R} - \frac{\mu_0 I}{20R}$$

$$\vec{B}_{net} = \frac{3\mu_0 I}{20R}\otimes$$

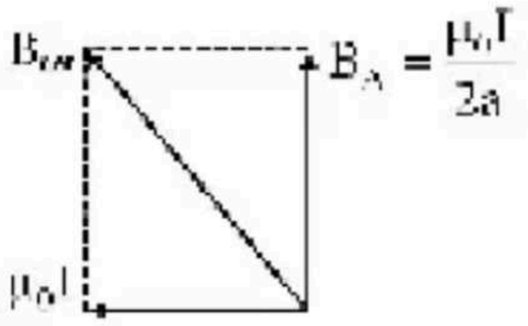
17) $B_C = \frac{\mu_0 I}{2r}, B_a = \frac{\mu_0 I r^2}{2(x^2 + r^2)^{3/2}}$ At $x = \frac{r}{2}$

$$B_a = \frac{\mu_0 I r^2}{2\left(\frac{r^2}{4} + r^2\right)^{3/2}}$$

$$= \frac{\mu_0 I r^2}{2\left(\frac{5}{4} r^2\right)^{3/2}} = \frac{\mu_0 I}{2r}\left(\frac{4}{5}\right)^{3/2}$$

$$= \frac{\mu_0 I}{2r}\left(\frac{2}{\sqrt{5}}\right)^3$$

18) The wire are kept perpendicular to the plane of the paper and currents are flowing out of the plane of the paper (say z-axis) then the magnetic field at point on the ± x-axis will be directed along
± y-axis. Its value at the middle of the two wires (origin of our co-ordinate system) is zero because the magnetic fields at O due to these wires are equal and oppositely directed. However, at a point to the right of wire 1, the magnitude resultant field B is not zero. Its value decreases from x = 0 to R/2 and similar is the case at points to the left of wire 2.
Similarly, the magnetic field at points to the left side of wire 1 and right side of the wire 2 falls rapidly with the increase of ± x but being the parallel currents in the wires, the directions of magnetic fields are oppositely directed hence the choice (D) is correct.

19) 

$$\therefore B_{net} = \frac{\sqrt{2}\mu_0 I}{2a}$$

20) $B = \mu_0 i / 2R$

$$i = \frac{2BR}{\mu_0}$$

$$m = iA = \frac{2BR\,\pi R^2}{\mu_0} = \frac{2\pi BR^3}{\mu_0}$$

21) $G = \frac{RS}{R - S}$

Case-1 $G_1 = \frac{3300 \times 80}{3220} = 81.2\Omega$

Case-2 $G_2 = \frac{5000 \times 80}{4920} = 81.2\Omega$

$\frac{G_1 + G_2}{2} \Rightarrow 81\Omega \simeq 80\Omega$

22)

By experiment

23) $B = \frac{\mu_0 i}{2\pi r} = \frac{2 \times 10^{-7} \times 50}{2.5}$

$= 4 \times 10^{-6}$ T

24)

Using $B = \frac{\mu_0 I}{2R}\left(\frac{\theta}{2\pi}\right)$; $B_{net} = B_1 + B_2$ & $\theta = \pi/2$

$B_{net} = \frac{\mu_0 I}{8}\left(\frac{1}{R_1} + \frac{1}{R_2}\right)$

25) $B = \frac{\mu_0 I}{2\pi r}\left(\frac{r^2 - R_{inner}^2}{R_{outer}^2 - R_{inner}^2}\right)$

$R_{inner} = R$

$R_{outer} = 2R$ $r = \frac{5R}{4}$

$B = \frac{\mu_0 I}{\frac{2\pi 5R}{4}}\left[\frac{\frac{25R^2}{16} - R^2}{4R^2 - R^2}\right]$

$\frac{4\mu_0 I}{2\pi(5R)}\left(\frac{9R^2}{48R^2}\right)$

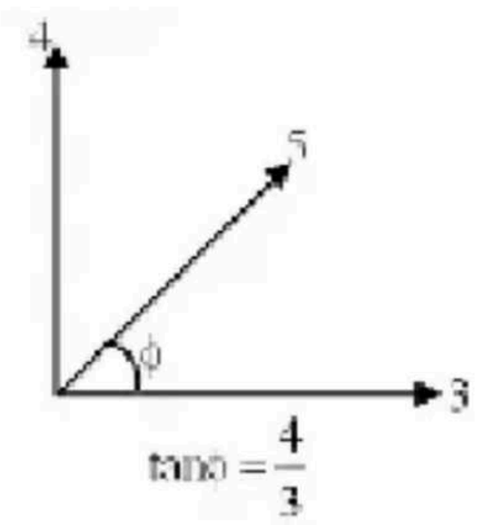
$\frac{2\mu_0 I}{5\pi R}\left(\frac{3}{16}\right)$

$\frac{3}{40}\frac{\mu_0 I}{\pi R}$

26) Force on the part of wire which is parallel to $\vec{B}$ will be zero. The forces on the other two parts will each be I$\square$B along mutually perpendicular directions.

27) $V^2 = V_R^2 + (V_L - V_C)^2$

Since $V_L = V_C$ hence $V = V_R = 200V$



28)

$i_3 = i_1 + i_2$

$= 3\sin\omega t + 4\cos\omega t$

$= 5\sin(\omega t + \phi)$
$= 5\sin(\omega t + 53°)$

29) Band width $= 232 - 212 = \frac{R}{L}$

$\therefore L = \frac{5}{20} = \boxed{250}\,mH$

30) Current in capacitor $I = \frac{V}{X_C}$

$I = (V) \times (\omega C)$

$C = \frac{I}{V\omega} = \frac{6.9 \times 10^{-6}}{230 \times 600} = 50pF$

31) if R = 0, P = 0

32) Capacitance of line = 0.01 × 300 μF
At resonance impedance is minimum

So $\omega L = \frac{1}{\omega C}$

$L = \frac{1}{\omega^2 C}$ [solve it]

33) From graph current Leads voltage
so possible components R–C
For R–C circuit

$\tan\phi = \frac{X_C}{R} \Rightarrow \frac{X_C}{R} = \tan \pi/4 = 1$

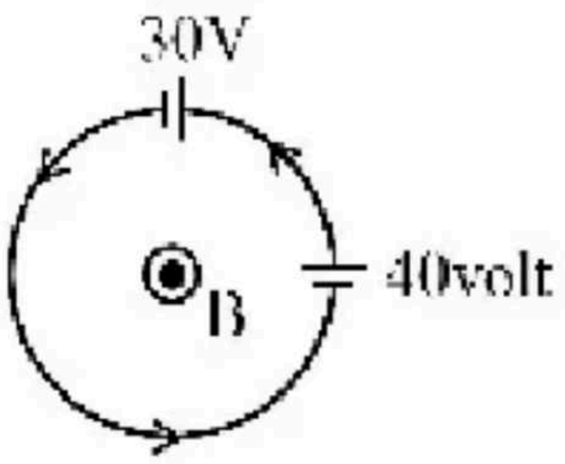
$X_C = R \Rightarrow \frac{1}{\omega C} = R \Rightarrow \frac{1}{100C} = R$

If R = 1kΩ then C = 10μF

34) ∴ R = 0
∴ $P_{av} = 0$

35)

Due to change in flux
e.m.f will be induced so net e.m.f



$$|e| = \frac{AdB}{dt}$$

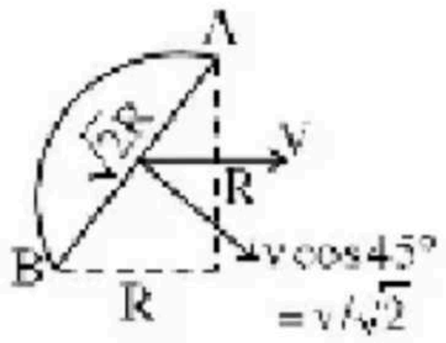
$|e| = 2 \times 20$

$= 40$ volt

Now

$$I = \frac{V}{R} = \frac{40-30}{10} = 1A$$

Anticlockwise



36)

$$e = B\left(\frac{V}{\sqrt{2}}\right)\sqrt{2}R$$

$= BvR$

37)

Magnetic force will act in upward direction on the loop while entering as well as exiting magnetic field. No magnetic force is there when loop is completely inside the magnetic field as flux is constant.

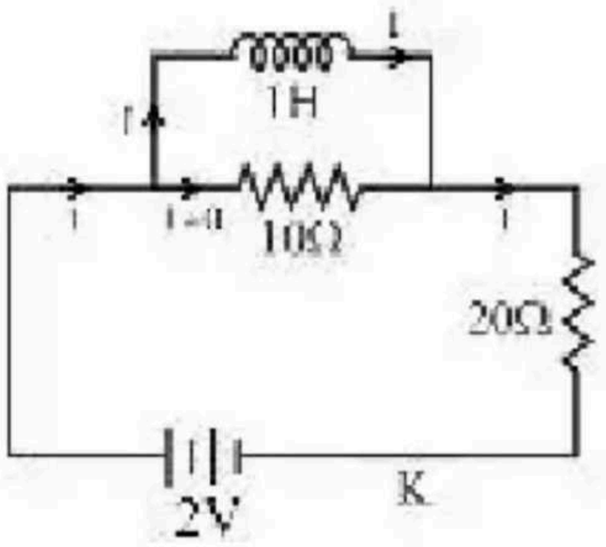
38)

$\phi = Mi = BA$

$$\Rightarrow Mi = \frac{\mu_0 i}{2b}\pi a^2$$

$$\therefore M = \frac{\mu_0 \pi a^2}{2b}$$

39) Long after the key is inserted, the circuit reaches steady state. In steady state with DC, an inductor behaves like a short circuit.



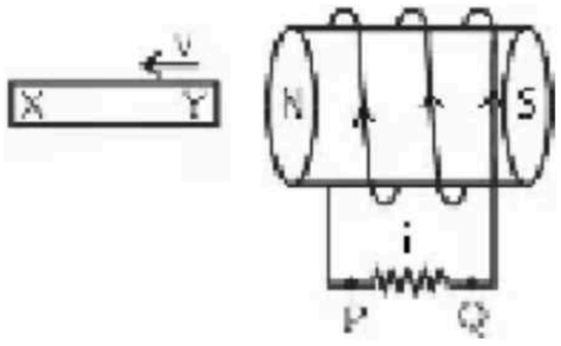
$I_{10\Omega} = 0$

40) $E \times 2\pi r = \pi a^2 \frac{dB}{dt}$

$$E = \frac{a^2 dB/dt}{2r} = \frac{\left(5 \times 10^{-2}\right)^2 \times 4}{2 \times 10 \times 10^{-2}}$$

$$E = \frac{25 \times 10^{-4} \times 4}{2 \times 10^{-1}} = \frac{100}{2} \times 10^{-3} = \frac{10^{-1}}{2} = 0.05 \text{ V/m}$$

41)



From figure given induced current in solenoid is flowing such that face of solenoid in front of bar magnet behave as 'N' and bar magnet is moveing away from solenoid so according to Lenz's law Y must be S pole.

42) $|e| = \frac{d\phi}{dt} = B\frac{dA}{dt} = B\frac{d}{dt}(\pi r^2) = 2\pi Br\frac{dr}{dt}$

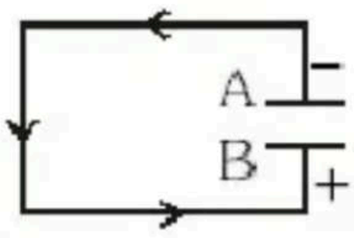
i.e. $e \propto r\frac{dr}{dt}$

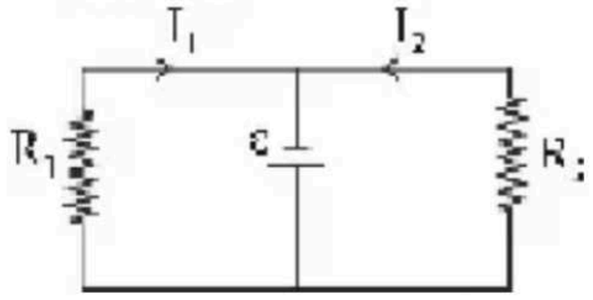
t = 0 – 1s, $\frac{dr}{dt} = 0$ so e = 0

t = 1–2 s, $\frac{dr}{dt}$ = constant so $e \propto r$

t > 2s, $\frac{dr}{dt} = 0$ so e = 0

43) t ↑ B ↑ then induced emf try to oppose





44)

45)

Source of time varying magnetic field may be

→ accelerated or retarded charge which produces varying electric and magnetic fields.

→ An electric field varying linearly with time will not produce variable magnetic field as

$$B_{ind} \propto \frac{dE}{dt} = \text{constant}$$

CHEMISTRY

46) (A) Down the group; radius increases
(B) EN does not decrease gradually from C to Pb
(C) Correct
(D) Correct
(E) Range of oxidation state of carbon ; –4 to +4

47) The ΔH_{eg} of oxygen is lowest among group 16 elements.

48)

$$\underset{\text{O.A (R.A)}}{\overset{+4}{PbBr_4}} \longrightarrow \overset{(+2)}{PbBr_2} + Br_2$$

$$\underset{\text{O.A (R.A)}}{\overset{+4}{PbI_4}} \longrightarrow \overset{(+2)}{PbI_2} + I_2$$

* In penta halides of Bi only BiF_5 exist.
* MnF_7 does not exist due to crowding

49)

NCERT XI, Pg. 314

50) In phosphonic acid or ortho phosphorous acid H_3PO_3 two P – OH, one P – H, one P = O bond is present.

51)

$SF_4 + 3H_2O \rightarrow H_2SO_3 + 4HF$
$P_4O_6 + 6H_2O \rightarrow 4H_3PO_3$
$PCl_3 + 3H_2O \rightarrow H_3PO_3 + 3HCl$
and SF_6 Hydrolysis not possible

52) NCERT Pg. # 181

53)

NCERT XI, Pg. # 318

54)

NCERT XII, Pg. # 181,182

55)

NCERT XII, Pg. # 192

56) NCERT Pg. # 216
ΔH_{eg} = Ne > Ar ≃ Kr > Xe >He

57) **Solution/ Explanation:**
The noble gases are generally inert in nature. Few noble gases is Xe and Rn shows some reactivity and that two with high electronegative element fluorine.
The only compound of Radon (Rn) which has been identified by radiotracer technique is RnF_2 (Radon difluoride)

58) **Solution:** Thorium (Z = 90) has ground state [Rn] $6d^2$ $7s^2$. Removing three electrons (first from 7s, then 6d) gives Th^{3+} = [Rn] $5f^1$.
Final Answer: Option (2)

59) Strongest oxidising agent = Co^{+3}
No. of unpaired e^- in $Co^{+3}[3d^6]$ = 4
Hence $\mu = \sqrt{n(n+2)} = \sqrt{24}\,BM$

60) (ii) $Cu^{+}_{(aq)} \rightarrow Cu^{+2}_{(aq)} + Cu_{(s)}$
(iii) $2Cr^{+2}_{(aq)} + 2H^{+} \rightarrow 2Cr^{+3}_{(aq)} + H_{2(g)}\uparrow$
(v) Zn, Cd, Hg are not consider as transition elements.

61) NCERT Pg. No.90

62) **Solution:**
In general the electronic configuration of outer orbitals of d-block elements is $(n-1)d^{1-10}$ ns^{1-2} except Pd which has $(n-1)d^{10}$ ns^0.
an orbital has maximum 2 electron.
Final Answer: The correct option is (2).

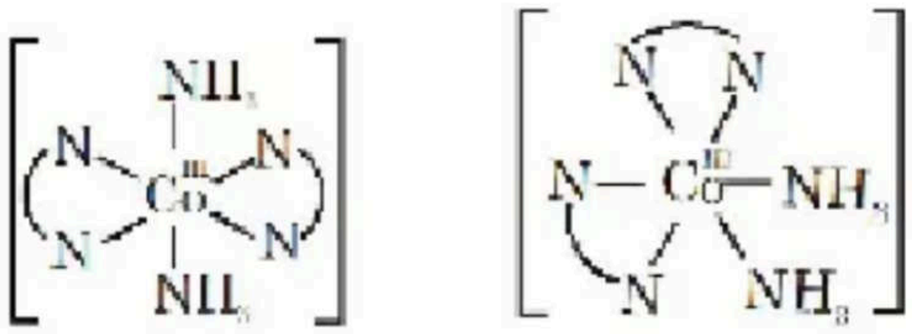
63) Order of basic strength in hydroxides of lanthanoids:-
$La(OH)_3$ > $Ce(OH)_3$ > $Pr(OH)_3$ > .....> $Lu(OH)_3$

64) NCERT Pg. NO.113

65) The magnetic properties of the actinoids are more complex than the lanthanoids.

66)

mabcd 2 cis 1 trans



trans optically inactive Cis (Optically active)

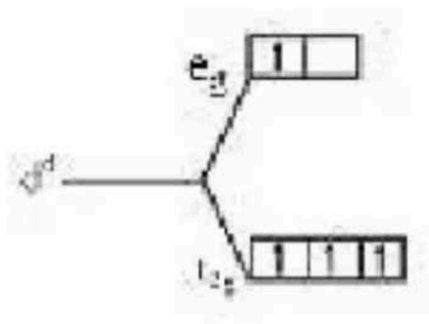
67)
$Co^{+3} \rightarrow [Ar], 3d^6$
Hybridisation d^2sp^3 in presence of strong field ligand.
Total 15 N–Co–N bond angles

68) NCERT Pg. No.119

69) NCERT Pg. No.130

70) NCERT Pg. No.132
Ni^{+2}, Pt^{+4}, Cr^{+3}

71) $[Fe(H_2O)_6]^{+2} \rightarrow$ Green colour solution due to d-d transition.
$[Ni(CN)_4]^{-2} \rightarrow$ is diamagnetic and it is colourless.
NCERT-12th, Pg. # 103, part-1



72)
back pairing is not possible because $p > \Delta_o$.

73)

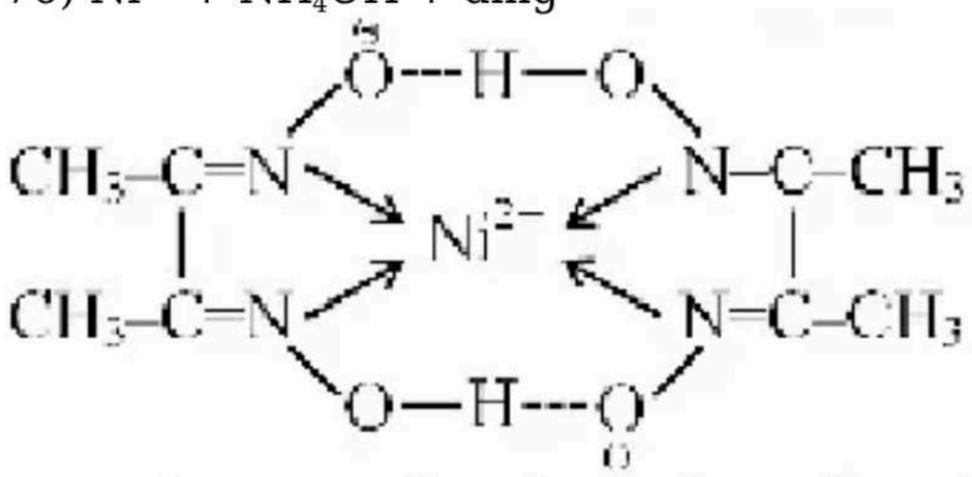
Ziegler catalyst → Titanium
Blood pigment → Iron
Wilkinson catalyst → Rhodium
Vitamin B_{12} → Cobalt

74) K_2MnO_4
$2 + x - 8 = 0$
$\Rightarrow x = +6$

O.S. of Mn = +6
IUPAC Name =
Potassium tetraoxidomanganate(VI)

75) $[Cr(H_2O)_6]^{3+}$ Contains $Cr^{3+} : [Ar]3d^3 : t_{2g}^3 e_g^o$
$[Fe(H_2O)_6]^{3+}$ Contains $Fe^{3+} : [Ar]3d^5 : t_{2g}^3 e_g^2$
$[Ni(H_2O)_6]^{2+}$ Contains $Ni^{2+} : [Ar]3d^8 : t_{2g}^6 e_g^2$
$[V(H_2O)_6]^{3+}$ Contains $V^{3+} : [Ar]3d^2 : t_{2g}^2 e_g^o$

76) $Ni^{2+} + NH_4OH + dmg \rightarrow$



2 Five member ring

III II
$Fe_4[Fe(CN)_6]_3$

Prussian Blue

77) $Na_2S + H_2SO_4 \rightarrow Na_2SO_4 + H_2S$
$(CH_3COO)_2Pb + H_2S \rightarrow PbS + 2CH_3COOH$
Black lead sulphide

78)

All the following metal ions will respond to flame test.
$Sr^{2+}, Ba^{2+}, Ca^{2+}, Cu^{2+}$

79) NCERT-XII, Part-1, Chapter-5, Pg. # 134

80) Nessler's Reagent Reaction :
$2K_2HgI_4 + NH_3 + 3KOH \rightarrow HgO.Hg(NH_2)I + 7KI + 2H_2O$
(Nessler's Reagent) (Iodine of Millon's base, Brown precipitate)

81)

Order of solubility : $Na_2S > ZnS > CuS$

82) H_2S + oxidizing agent → S (yellow turbidity)

83) Hyroxides of A^{3+}, Fe^{3+} and Cr^{3+} are insoluble in NH_3

84) **Solution:** A brick-red color imparted to a Bunsen flame is the characteristic test for a calcium (Ca^{2+}) salt.
Final Answer: Option (1) Ca salt.

85) **Solution:** The borax bead test is characteristic for many transition metals that form colored metaborates. Zn^{2+} has a filled d-shell (d^{10}) and forms a colorless (white when cold) bead, and is therefore considered not to give a positive test in the typical sense.
Final Answer: Option (4) Zn^{2+}.

86) Stabiliby of complex α stability constant

87) $[Co(NH_3)_5\ NO_2]Cl$ and $[Co(NH_3)_5ONO]Cl$
NO_2 is ambidentate so linkage isomers.
$[Cu(NH_3)_4]\ [PtCl_4]$ and $[Pt(NH_3)_4]\ [CuCl_4]$ for coordination isomerism cation and anion must be complex.
$[Pt(NH_3)_4\ Cl_2]\ Br_2$ and $[Pt(NH_3)_4\ Br_2]Cl_2$
Out side Co-ordination sphere is having different ionisable spescis that's why it shows ionisation isomerism.

88) (i) $[Ag(NH_3)_2]^+$ – sp linear
(ii) $MnCl_4^{2-}$ – sp^3 Tetrahedral
(iii) $[Cu(NH_3)_4]^{+2}$ – dsp^2 square planar
(iv) $Ni(CN)_4^{2-}$ – dsp^2 square planar

89) $CuSO_4.5H_2O$ is $[Cu(H_2O)_4]SO_4.H_2O$
4, water molecules are attached as secondary valency and 1, water molecule is attached due to H-bond.

90)
$[Co(en)_2Cl_2]^+$ has 2 optically active and 1 optically inactive isomer.

BIOLOGY

91) NCERT-XII, Pg # 61

92) NCERT-XII, Pg # 62

93) NCERT-XII, Pg # 66

94) NCERT XII Pg. No. # 63

95) NCERT XII Pg. No. # 57, 60, 63

96) NCERT XII Pg. No. # 64, 65, 77

97) XII NCERT Pg. No. 57, 64

98) XII NCERT Pg. 60, 61

99)

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100) NCERT XII, Page No. 73(E), 81(H)

101) NCERT XII, Page No. 73, 74(E), 82(H)

102) NCERT XII, Page No. 55, 62 (E), 62, 70 (H)

103) NCERT XII, Page No. 71

104)

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105) NCERT XII, Pg. # 58/64

106) NCERT, Pg. # 56/63

107)

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108) NCERT XII, Page No.: 70,71

109) NCERT XII, Pg # 62,69-70 (E)
Pg # 69,76-77 (H)

110) NCERT-XII, Pg#76

111) NCERT XII, Page # 72, 73

112) NCERT XII, Page # 73

113) NCERT-XII, Pg # 106

114) NCERT-XII, Pg # 97

115) NCERT-XII, Pg # 105, 106

116) NCERT-XII, Pg # 99

117) NCERT-XII, Pg # 89

118) NCERT-XII, Pg # 81

119) NCERT-XII, Pg # 92

120) NCERT XII, Pg. # 92-93/103

121) NCERT XII, Pg. # 84-86/93-95

122) NCERT XII, Pg. # 83, 84

123) NCERT (XII) ; Page No. # 89

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125) NCERT (XII) ; Page No. # 81

126) NCERT XII, Pg. # 93

127) NCERT XII, Pg # 91

128) NCERT XII, Pg. # 93/103

129) NCERT XII, Pg. # 106/116, 117

130) NCERT-XII, Pg # 91

131) NCERT-XII, Pg # 90

132) NCERT-XII, Pg # 89

133) NCERT-XII, Pg # 82

134) NCERT-XII, Pg # 96

135) NCERT-XII, Pg # 94

136)

NCERT Pg.#171

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NCERT XII, Pg. # 168

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NCERT XII, Pg. # 168

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NCERT XII, Pg. # 168

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NCERT XII, Pg. # 165

142) NCERT Pg. # 170

143) NCERT 12th, Page No. # 181, 182

144) NCERT Pg. # 169

145) NCERT Pg.#173.

146) NCERT 12th, Page No. # 180,183,184

147) Old NCERT-XII, Pg. # 170

148) NCERT Pg. # 165

149) NCERT Pg. # 173

150) NCERT-XII, Pg. # 181, 182

151) NCERT 12th, Page No. # 171
A,C,D,E,F are not used

152) NCERT 12th, Page No. # 170-171

153) NCERT Pg.#170,173,174,185,168

154) NCERT-XII Pg # 182

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158)

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160) NCERT 12th, Page No. # 184

161)

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162) NCERT Pg. # 184,171

163) NCERT 12th, Page No. # 180

164) NCERT-XII, Pg. # 184, 185

165) NCERT-XII, Pg. # 179 (E), 196 (H)

166) NCERT 12th, Page No. # 183

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169) NCERT Pg. # 153 (E), 168-169 (H)

170) NCERT XII Page No. 182

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172) NCERT-XII, Pg. # 152, 153 (E), 167, 168, 169 (H)

173) NCERT Page NO. 154

174) NCERT Pg. # 153 (E), 168 (H)

175) NCERT Pg. # 157 (E), 173 (H)

176) NCERT Page No. 154

177) NCERT Page No. 158

178) NCERT Page No. 151, 152, 153

179) NCERT-XII, Pg. # 150

180)

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